Model Augmentation for a Dual-Gantry High-Precision Motion System

Dirk van den Berg
Control Systems Group, Dept. of Electrical Engineering
Eindhoven University of Technology
Eindhoven, The Netherlands

***Abstract*—**

- **Problem: improve settling prediction for a position-dependent dual gantry without surrendering the meaning of its physical parameters**
- **Method: combine a quasi-LPV first-principles baseline, dynamic parallel LFR augmentation, closed-loop rollout training, and state-level orthogonal projection**
- **Evidence: compare baseline, unconstrained augmentation, orthogonal variants, and a black-box model on the completed validation ladder, including relevant motion and settling where available**
- **Main result: report predictive improvement together with the measured limits from short-horizon training, inactive added states, and non-orthogonal true missing dynamics**
- **Conclusion: state when the proposed decomposition is useful and which conditions prevent interpreting it as true parameter recovery**

[TODO: Write the numerical abstract only after the final simulation and Telica result tables are frozen.]

***Index Terms*—**

I. Introduction

ASMPT develops semiconductor assembly and packaging equipment, including high-precision motion systems for advanced packaging, which combines multiple semiconductor functions within a single package. Such technologies place strict demands on alignment accuracy, yield, and manufacturing cost [1], [2].

This paper focuses on a dual-gantry motion system in which two parallel linear actuators support a common beam carrying the payload. The shared beam couples the actuator dynamics, so motion of one actuator influences the other, and the system exhibits position-dependent behaviour [3], [4] and flexible dynamics [3]. In such systems, settling behaviour is a key performance metric. After each motion, the system must wait for residual oscillations to decay before the next placement can begin [5], so settling time directly affects throughput. Furthermore, cross-coupling means that the motion of one gantry can disturb the other, making accurate prediction relevant not only for settling but also for tracking behaviour during simultaneous motion.

Predicting this behaviour therefore requires an accurate representation of the underlying plant dynamics. Such a model can be used as a digital twin, for example to evaluate a changed controller before it is tested on the machine. Digital twins can support offline simulation of semiconductor manufacturing equipment and help optimize operating parameters [6]. Such models are also valuable for control design and for process and root cause analysis. This requires a model with physically meaningful parameters, because a change in machine behaviour can then be traced back to a physical cause, such as a change in stiffness or damping.

White-box models are attractive because they reflect the physics of the system, which makes them interpretable. First-principles (FP) models are a specific type of white-box model, derived from physical laws and described by quantities such as mass, stiffness, and damping. However, purely physics-based models do not always capture all behaviour needed for accurate prediction across the operating range [7]. More accurate FP models can be developed, but this is labour-intensive [8].

Purely data-driven (black-box) models such as neural networks or Gaussian processes can describe measured behaviour well, but they give less physical insight and are often harder to trust at operating conditions not covered by the training data [7]. Typically these models also require large amounts of data to train [7], [9]. In addition, these models need to learn behaviour that is already known from first principles, such as the rigid-body dynamics [8]. Physics-informed approaches, including physics-informed neural networks (PINNs), enforce physical laws as constraints during training [9], [10]. However, these methods require the physics equations to be known and still rely on a black-box model for the entire system [8]. For the plant model of the dual-gantry system, both predictive accuracy and physical interpretability are required. Physical interpretability means that the estimated physical parameters are unique [11]. [**TODO:** Leave out?: Section **??** gives the conditions under which they equal their true values.]

However, neither a purely physics-based nor a purely data-driven model is sufficient on its own [7]. Grey-box modeling is the logical choice, because it is able to combine a physically meaningful baseline model needed for interpretation with data-driven flexibility needed for accurate prediction [12]. The following grey-box approaches appear in practice: parameter re-estimation within the physics model structure, static residual correction on the model output, and dynamic augmentation that extends the baseline with learned states [7], [8]. Model augmentation combines a physics-based baseline model with a learned component, so that the learned component only needs to capture the dynamics the physics-based model cannot describe [8]. However, related work on an ASMPT wire bonder shows that preserving physical interpretability during model updating

remains an open challenge [13].

Learning-based hybrid modeling has already been demonstrated on real industrial systems [14]. However, a key gap remains: there is no grey-box modeling framework for the dual-gantry system that simultaneously addresses coupled multi-input multi-output (MIMO) dynamics, position-dependent behaviour, closed-loop data, and preserves physical interpretability. Model augmentation using a linear fractional representation (LFR) has already been demonstrated, but not for position-dependent baselines or closed-loop data [8]. The related work at ASMPT uses closed-loop data, but keeps the physical parameters fixed during training [13]. Orthogonal-by-construction augmentation is able to preserve the physical meaning of the baseline parameters, but has so far only been formulated for discrete-time input-output models [11]. This paper therefore aims to develop and evaluate such a framework in simulation, so that the plant model becomes more accurate across operating points and motion profiles while remaining physically meaningful.

This paper addresses the following research question: *For the dual-gantry system, how should a physics-based plant model be augmented to capture unmodelled dynamics, improving prediction accuracy across operating points and motion profiles while preserving the physical interpretability of its parameters?* To answer this research question, the following four aspects are investigated:

1) In what form should the baseline model be represented for subsequent augmentation, and does a position-dependent (LPV) formulation improve prediction compared to position-independent (LTI) alternatives?
2) What augmentation structure can effectively capture dynamics not modeled in the physics-based baseline model?
3) How can parameter interpretability be maintained during data-driven augmentation, compared to unconstrained approaches?
4) How well does the augmented model generalize to (i) held-out operating points and (ii) unseen motion profiles, and how does its prediction accuracy compare to a black-box state-space model?

[**TODO:** Confirm whether references and appendices are excluded from the 12-page IEEE limit and whether TU/e requires front matter beyond the IEEE paper.]

II. Dual-Drive Gantry System and Physics Baseline

A. System and first-principles baseline

[**TODO:** state we leave out coulomb friction and that we will use for the experiment in the simulation in section V] The physics baseline of the gantry system used in this paper is the lumped-parameter model of the dual-drive gantry of García-Herreros et al. [3]. The gantry has two actuators that are connected with a cross-arm and a third actuator that moves the payload along this arm (Fig. 1). [**TODO:** Does this make sense to put here?: Dynamics not represented by the baseline are treated as model discrepancy and addressed by the augmentation in Section III.] Define the actuator forces $u_{\text{act}} = [F_{X1}\ F_{X2}\ F_Y]^\top$, the measured positions $q_{\text{act}} = [X_1\ X_2\ Y]^\top$, and the generalized coordinates $q = [X\ \Theta\ Y]^\top$. Let L_b denote the cross-arm length between the two flexible joints. The coordinate and force transformations are

$$q = P^{-\top} q_{\text{act}}, \qquad u = P u_{\text{act}}, \\ P = \begin{bmatrix} 1 & 1 & 0 \\ L_b/2 & -L_b/2 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \tag{1}$$

where u denotes the generalized force. The generalized coordinates are defined as $X = (X_1 + X_2)/2$ and $\Theta = (X_1 - X_2)/L_b$, while Y is unchanged. These coordinates avoid [**TODO:** This needs to make clear why we choose for this realization. a closed kinematic-chain description and make the coupling caused by the nonuniform load distribution explicit] [3]. [**TODO:** How to connect, nonunfirom ... moving payload position] Moving the payload position Y changes the mass distribution of the gantry and therefore its inertia. This position dependence is retained by representing the baseline in linear parameter-varying (LPV) form, with the measured payload position Y as scheduling variable [15].

Following García-Herreros et al. [3], the baseline uses $\cos\Theta \approx 1$ and $\sin\Theta \approx 0$ and assumes that the Coriolis and centripetal force term is negligible compared with the inertia, damping, and stiffness forces. Under these assumptions, the baseline equations of motion are

$$M(Y)\ddot{q}(t) + C\dot{q}(t) + Kq(t) = P u_{\text{act}}(t), \tag{2}$$

with

$$M(Y) = \begin{bmatrix} m_1 + m_2 + m_b + m_h & \frac{L_b}{2}(m_1 - m_2) - m_h Y & 0 \\ \frac{L_b}{2}(m_1 - m_2) - m_h Y & J_b + J_h + \frac{L_b^2}{4}(m_1 + m_2) + m_h(d^2 + Y^2) & -m_h d \\ 0 & -m_h d & m_h \end{bmatrix}, \tag{3}$$

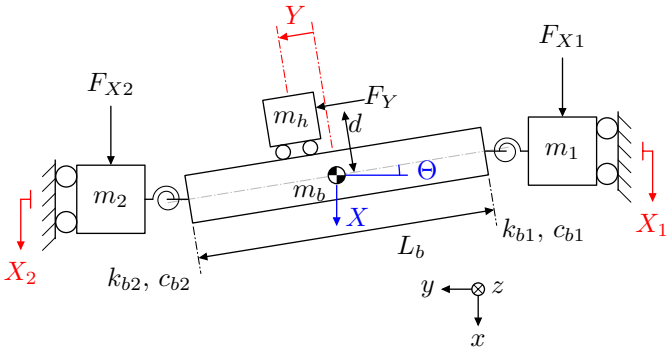


Fig. 1: Still need to add springs to the current torsion. [**TODO:** Lumped-parameter gantry schematic adapted from the mechanical arrangement in [3]. Red labels denote actuator coordinates and blue labels generalized coordinates.]

[**TODO:** Will need to add springs to the current torsion; feedback from Jasper]

$$C = \begin{bmatrix} c_{g1}+c_{g2} & \frac{L_b}{2}(c_{g1}-c_{g2}) & 0 \\ \frac{L_b}{2}(c_{g1}-c_{g2}) & c_{b1}+c_{b2}+\frac{L_b^2}{4}(c_{g1}+c_{g2}) & 0 \\ 0 & 0 & c_y \end{bmatrix},$$
$$K = \begin{bmatrix} 0 & 0 & 0 \\ 0 & k_{b1}+k_{b2} & 0 \\ 0 & 0 & 0 \end{bmatrix}. \quad (4)$$

[**TODO:** Proposed insertion: "García-Herreros et al. also model Coulomb friction at the three actuators [3, Sec. 2.2]. The baseline omits these terms, so that friction remains a physical effect the augmentation must capture (Section V)." Reason documented in D-204. Second reason? A sign term is not twice differentiable, which [8, Cond. 8] assumes for well-posedness. because: ?] [**TODO:** Look at this section, the above state a lot, look fine] The masses, flexible-joint stiffness and damping coefficients, and [**TODO:** geometric dimensions] are shown in Fig. 1. Here, J_b and J_h denote the yaw inertias, while c_{g1}, c_{g2}, and c_y denote the actuator damping coefficients. The nominal values are taken from [3, Table I] and listed in Appendix A-C [**TODO:** Not sure if to add this, depends on real data or not; they were not identified on the present Telica machine.] To prevent the definition of q from being parameter-dependent, the cross-arm length L_b is fixed at the nominal value listed in Appendix A-C rather than treated as a free model parameter. [**TODO:** Determine and justify the operating range of Y; distinguish the mechanical stroke, simulated-data range, and range covered by the measured ILC records.]

B. Self-scheduled LPV-LFR representation

- Y is a state and the scheduling variable; $M(Y)^{-1}$ makes the model rational in Y (Drenth)
- $M(Y) = M_0 + YM_1 + Y^2M_2$; Y enters only through $Y\ddot{q}$ and $Y^2\ddot{q}$, realized by repeating Y (this work)
- Latent variables z, w and Delta-block $\Delta(Y) = YI_6$; not claimed minimal (this work)
- LTI part G in the continuous-time form of Drenth Eq. (6); its first block row is the state equation (adapted)
- Closing the algebraic loop gives back (2) exactly (this work)
- Well posed iff $M(Y)$ nonsingular, true for every Y; closed-form $M(Y)^{-1}$, no numerical solve (Drenth Def. 1; this work)

The measured payload position Y is both a state and the scheduling variable, making the baseline a self-scheduled LPV model [**TODO:** source, or do i state it earlier..?]. Solving (2) for $\ddot{q}$ requires $M(Y)^{-1}$, so the model is rationally dependent on Y. An Linear Fractional Representation (LFR) is introduced to separate this rational scheduling dependence. We move the dependence out of the LTI part G and into a Delta-block $\Delta(Y)$, with the scheduling being connected in feedback with G (Fig. 2) [8], [16].

To separate the scheduling variable Y, deconstruct the inertia matrix (3) into its Y dependent components,

$$M(Y) = M_0 + YM_1 + Y^2M_2. \quad (5)$$

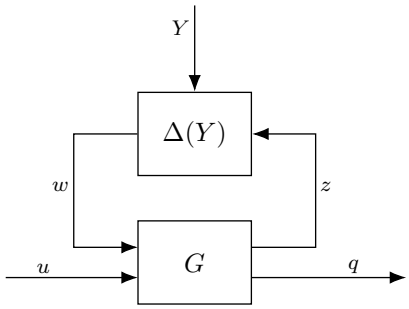


Fig. 2: [**TODO:** Self-scheduled baseline interconnection. The Delta-block closes the latent variables of G through $w = \Delta(Y)z$, with $\Delta(Y) = YI_6$ and Y a component of the state.]

M_0, M_1 and M_2 are given in Appendix A-A. Substituting (5) into (2) gives

$$M_0\ddot{q} + M_1(Y\ddot{q}) + M_2(Y^2\ddot{q}) = u - C\dot{q} - Kq. \quad (6)$$

Here Y only enters through $Y\ddot{q}$ and $Y^2\ddot{q}$, with $Y^2\ddot{q} = Y(Y\ddot{q})$. This lets us apply the same scheduling variable twice instead of introducing a second scheduling variable Y^2,

$$\ddot{q} \xrightarrow{Y} a_1 \xrightarrow{Y} a_2, \quad (7)$$

with $a_1 = Y\ddot{q}$ and $a_2 = Y^2\ddot{q}$.

We treat a_1 and a_2 as independent signals (6) becomes

$$M_0\ddot{q} + M_1a_1 + M_2a_2 = u - C\dot{q} - Kq, \quad (8)$$

which contains no Y. The latent variables are $z = \mathrm{col}(\ddot{q}, a_1)$ and $w = \mathrm{col}(a_1, a_2)$, with the Delta-block mapping z to w.

$$w = \begin{bmatrix} a_1 \\ a_2 \end{bmatrix} = \begin{bmatrix} Y\ddot{q} \\ Ya_1 \end{bmatrix} = \underbrace{\begin{bmatrix} YI_3 & 0 \\ 0 & YI_3 \end{bmatrix}}_{\Delta(Y)=YI_6} \begin{bmatrix} \ddot{q} \\ a_1 \end{bmatrix} = \Delta(Y)z. \quad (9)$$

We do not claim this realization to be minimal [**TODO:** I have an SVD reduction add this in the appendix or not?]. We can write the LTI part G of the LFR in the continous-time form, [16, Eq. (6)],

$$\begin{bmatrix} \dot{\tilde{x}} \\ z \\ q \end{bmatrix} = G \begin{bmatrix} \tilde{x} \\ w \\ u \end{bmatrix}, \quad (10)$$

where $\tilde{x} = \mathrm{col}(q, \dot{q})$ is the state and q the output. The first block row is the state equation, with $\ddot{q}$ obtained from (8) by multiplying with M_0^{-1}. For the blocks of G Appendix A-A. Closing the algebraic loop with (9) results in $a_1 = Y\ddot{q}$ and $a_2 = Y^2\ddot{q}$, which turns (8) back into (2). The LFR is therefore an exact realization of the baseline. The LFR is well posed if the algebraic loop has a unique solution for every Y [16, Def. 1]. For the gantry realization we show that this conditions holds when $M(Y)$ is nonsingular (Appendix A-A). For positive masses and inertias, $M(Y) \succ 0$ for every $Y \in \mathbb{R}$ (Appendix A-D). Therefore the LFR is well posed over the whole Y-operating range. [**TODO:** Connect the two sentences better?]

C. Physical parameters and identifiable combinations

- With q measured, known force scale and full actuation, the input-output behaviour fixes M_0, M_1, M_2, C, K and conversely (this work)
- Twelve nonzero entries, three equal to m_h, leave ten independent functions: exactly ten combinations are structurally identifiable (this work; definition Ovchinnikov)
- Written physically as θ_{base}, the gantry counterpart of base parameters (Gautier and Khalil); one-to-one with the entries for $m_h > 0$ (this work)
- Four lost directions; estimates reported in θ_{base}; result independent of the encoder; augmented model in Section IV; recovery from the records in Setup
- Positive θ_{base} does not give $M(Y) \succ 0$; that holds iff (12) (this work; proof Appendix A-D)
- Training keeps (12) at every iterate; the map and its justification are in Section III-D

The fourteen parameters θ of [**TODO:** needs to have appendix in the name of the table] Table III cannot all be identified from input-output data. The positions X_1, X_2, Y are measured, which gives q through (1), and the forces u are known in newtons. Two parameter sets θ and θ' that produce the same motion for every input force satisfy

$$\big(M(Y) - M'(Y)\big)\ddot{q} + (C - C')\dot{q} + (K - K')q = 0$$

With unconstrained initial conditions and full actuation, any acceleration $\ddot{q}$ can be reached at any state within the operating range. Therefore the identity must hold for all these states and accelerations, this implies that $K = K'$, $C = C'$ and $M(Y) = M'(Y)$. Since $M(Y)$ is quadratic in Y, this also implies that $M_0 = M_0'$, $M_1 = M_1'$ and $M_2 = M_2'$. If these matrices are equal, the input-output behaviour is also the same. Therefore, the part of θ that can be identified is determined by the parameter combinations in M_0, M_1, M_2, C and K. If we ignore the symmetric entries, (3) and (4) contain twelve nonzero entries. Three of these, $M_{0,33}$, $-M_{1,12}$ and $M_{2,22}$, are equal to m_h. This leaves us with ten distinct functions of θ. The Jacobian of these ten functions with respect to $(m_h, d, m_1, m_b, J_b, c_{g1}, c_{g2}, c_{b1}, c_y, k_{b1})$ is block triangular and has a determinant of $m_h L_b^2/2 \neq 0$, meaning that the functions are independent. Since the matrices only depend on these ten functions, exactly ten independent combinations of θ can be identified [**TODO:** check source] [17, Def. 7].
We define these ten combinations as

$$\theta_{\text{base}} = [k_{b,\Sigma}, c_{g1}, c_{g2}, c_y, c_{b,\Sigma}, m_h, m_\Sigma, m_\Delta, J_{\text{eff}}, d]^\top, \quad (11)$$

with $k_{b,\Sigma} = k_{b1} + k_{b2}$, $c_{b,\Sigma} = c_{b1} + c_{b2}$, $m_\Sigma = m_1 + m_2 + m_b$, $m_\Delta = m_1 - m_2$ and $J_{\text{eff}} = J_b + J_h + \frac{L_b^2}{4}(m_1 + m_2)$. This follows the idea of base parameters used in robot dynamics [**TODO:** check source] [18]. For $m_h > 0$, θ_{base} can be uniquely determined from the ten entries (Appendix A-B). We therefore estimate θ_{base} instead of θ, presented in the results [**TODO:** Section **??**].

Positive θ_{base} does not imply $M(Y) \succ 0$. For $m_h, m_\Sigma, J_{\text{eff}} > 0$, $M(Y) \succ 0$ for every Y if and only if

$$m_\Sigma J_{\text{eff}} > \tfrac{L_b^2}{4}\, m_\Delta^2 \quad (12)$$

(Appendix A-D). To ensure that well-posedness and (12) hold during training, we use a logarithmic parameterization for the nine positive combinations and a $\tanh$-bounded m_Δ (Section III-D).

III. Dynamic Augmentation of the LPV-LFR Baseline

- The section extends the baseline with learned dynamics and identifies the result from closed-loop records; it names its five subsections (style profile)
- Contribution share: the realisation on the self-scheduled gantry baseline with joint estimation of θ_{base}, its identification in closed loop with the known controller, and the PS initialisation of the added states (MUST ESTABLISH)

The dynamics of the system that are not captured by the baseline model defined in Section II-B will be identified from input-output data measured in closed-loop. The baseline model will be augmented to represent these missing dynamics. A correction to the states of the baseline model alone is not sufficient, so additional states are introduced. We make use of a neural network to model both the changes for the baseline states and the additional states.
The model is trained using short windows of the closed-loop data, where each window is simulated with the known controller. The initial states are estimated by an encoder, including the additional states. During training, the physical parameters θ_{base} are also estimated, using the combinations defined in Section II-C.

A. Augmented model

- If the omitted effects are dynamic, the six baseline states cannot represent them; refitting θ_{base} or a static augmentation adds no states (EJC Sec. 2)
- Hoekstra's LFR augmentation structure with a fixed interconnection; the self-scheduled LPV-LFR is its baseline block; the learned block is a network, not Drenth's LPV-LFR augmentation (thesis Eq. (5.2)) (?: reason)
- Parallel because the orthogonal-by-construction method of Section IV is defined for the additive structure (Györök 2026); output map not augmented (Kessels Remark 5.3)
- The structure is a discrete-time state update (EJC Sec. 2, Table 1), so f_{base} is one RK4 step with the input held and $M(Y)$ per stage; the correction is added after the step
- Acyclic interconnection plus the baseline result of Section II: well posed for every Y (EJC Thm. 1)
- Normalisation: affine, statistics of the training records, velocities from differenced positions; later displays in normalised symbols
- One tanh MLP supplies f_{aug} and g_{aug} (EJC Sec. 4.3); it corrects every physical row, so a persistent correction on X or Y accumulates in position (?: justification); $n_{\bar{x}} = 0$ is the static parallel augmentation

- The baseline/network split is not unique (2026 Sec. 5.2); interpretable means the estimated θ_{base} keeps its physical meaning (Györök Sec. 3); Section IV makes the estimate unique

To represent additional dynamics such as a flexible mode, refitting θ_{base} or correcting the baseline states with a static augmentation is not sufficient. Both these methods [**TODO:** only retain] the original states of the baseline, while missing dynamics require states the baseline does not include [8, Sec. 2]. A dynamic augmentation therefore adds states $\bar{x}$ to model these missing dynamics [8, Sec. 2]. The augmented model updates these states using the full state and input [8, Sec. 2, Table 1]. The LFR-based augmentation structure of Hoekstra et al. is used, an interconnection matrix connects a baseline block and a learning block [8, Sec. 3.1]. The learning components are a neural network consisting of f_{aug} and g_{aug}. [**TODO:** constant block is not really constant] Similar to the LFR in Section II-B (Fig. 2), a constant matrix connects the two blocks through latent variables. The self-scheduled LFR The baseline block for this is the self-scheduled LFR of Section II-B, this is a nonlinear [**TODO:** Check can i state nonlinear here] map of the state and the input [19, Sec. 2.4]. The augmented model is thus an LFR whose baseline block is itself an LFR in Y.

The dynamic-parallel augmentation structure is chosen, because the orthogonal-by-construction parametrisation that Section IV builds upon is defined for the additive structure [11].

The output map is not augmented (no h_{aug}), because (1) gives the exact relation between the states and the measured outputs. This is also done to ensure that q still represents the physical positions. With a learned correction to the output map the states could stop representing the actual positions. Kessels also leaves the output map unchanged when it is already accurate [13, Remark 5.3]. [**TODO:** not an LPV-LFR with a learned scheduling map as in the augmentation of Drenth [19, Eq. (5.2)].] [**TODO:** Missing: reason for a network as learning block rather than Drenth's LPV-LFR augmentation. The research plan (Aspect 2) extends the structure "to the LPV setting following" Drenth and leaves "the specific parameterization to be established". Candidates: (a) Hoekstra's framework with its zero start and well-posedness result is kept as is; (b) the research question asks only for a structure that captures the omitted dynamics. Close D-017 with a decision entry.]

The model is defined in discrite time with the augmentation added to the state update of the baseline [8, Sec. 2, Table 1]. This gives us

$$\begin{aligned}\tilde{x}_{k+1} &= f_{\text{base}}(\theta_{\text{base}}, \tilde{x}_k, u_{\text{act},k}) + f_{\text{aug}}(\theta_{\text{aug}}, \hat{x}_k, u_{\text{act},k}),\\ \bar{x}_{k+1} &= g_{\text{aug}}(\theta_{\text{aug}}, \hat{x}_k, u_{\text{act},k}),\\ \hat{y}_k &= P^\top q_k,\end{aligned} \tag{13}$$

where $\tilde{x}_k = \text{col}(q_k, \dot{q}_k) \in \mathbb{R}^6$ is the physical state of (10), $\bar{x}_k \in \mathbb{R}^{n_{\bar{x}}}$ the added state, $\hat{x}_k = \text{col}(\tilde{x}_k, \bar{x}_k)$, $u_{\text{act},k} \in \mathbb{R}^3$ the actuator forces, $\hat{y}_k \in \mathbb{R}^3$ the predicted actuator positions, and $y_k = q_{\text{act},k}$ the measured positions, $\theta_{\text{base}} \in \mathbb{R}^{10}$ the combinations (11), and [**TODO:** using theta might be confusing with baseline] $\theta_{\text{aug}} \in \mathbb{R}^{n_{\text{aug}}}$ the network parameters. The added states are updated by g_{aug} without any predetermined physical meaning, in contrast to the baseline states $\tilde{x}$ for which the positions are measured.

The baseline is evaluated over one sample period T_s. Writing (2) in state-space form gives

$$\dot{\tilde{x}} = F(\tilde{x}, u) = \begin{bmatrix} \dot{q} \\ M(Y)^{-1}\big(u - C\dot{q} - Kq\big) \end{bmatrix}, \tag{14}$$

where $u = Pu_{\text{act}}$ is the generalized force (1), M, C and K depend on θ_{base} through (11), and $M(Y)^{-1}$ is the closed loop of the LFR of Section II-B. f_{base} is calculated using one fourth-order Runge-Kutta (RK4) step of F over T_s, with the input kept constant during the step. For each of the four RK4 steps $M(Y)$ is evaluated using the value of Y from its intermediate state.

[**TODO:** The augmented model admits a unique solution for every state, input and Y. The interconnection has no algebraic loops, making it acyclic, and both blocks are twice continuously differentiable, which together give well-posedness [8, Cond. 6, Cond. 8, Thm. 9]. The baseline LFR is also well posed (Section II-B).]

The inputs to the neural network are normalised to zero mean and unit variance, which has been shown to improve the estimation of models containing neural networks [8, Sec. 5.3]. The mean and standard deviations are computed from the training data. For the baseline states the measured positions are converted to their generalized positions, for which the velocities can be estimated from their finite difference by dividing with T_s. This ensures that the positions and outputs are scaled with the same measurements: keeping the scales consistent. This also makes sure that all models independent of its initial parameters use the same scale,

$$\tilde{x}^{\text{n}} = T_x(\tilde{x} - \mu_x), \quad u^{\text{n}} = T_u(u_{\text{act}} - \mu_u), \quad y^{\text{n}} = T_y(y - \mu_y), \tag{15}$$

where $\mu_x \in \mathbb{R}^6$ and $\mu_u, \mu_y \in \mathbb{R}^3$ are the channel means, and $T_i = \text{diag}(\sigma_i)^{-1}$ for $i \in x, u, y$, with σ_i the corresponding standard deviations. The additional states $\bar{x}$ are not normalised, so $\hat{x}^{\text{n}} = \text{col}(\tilde{x}^{\text{n}}, \bar{x})$.

After normalisation the model equations become

$$\begin{aligned} f^{\text{n}}_{\text{base}}(\theta_{\text{base}}, \tilde{x}^{\text{n}}, u^{\text{n}}) &= T_x\big(f_{\text{base}}(\theta_{\text{base}}, \tilde{x}, u_{\text{act}}) - \mu_x\big),\\ f^{\text{n}}_{\text{aug}} &= T_x f_{\text{aug}}, \qquad g^{\text{n}}_{\text{aug}} = g_{\text{aug}},\\ \hat{y}^{\text{n}}_k &= C^{\text{n}}\tilde{x}^{\text{n}}_k, \qquad C^{\text{n}} = T_y P^\top [I_3 \;\; 0]\, T_x^{-1}. \end{aligned} \tag{16}$$

where $\tilde{x}$ and u_{act} follow from $\tilde{x}^{\text{n}}$ and u^{n} by inverting (15), so RK4 uses the states and inputs in physical units. The correction represents a change in the states itself and not the full state itself, and therefore we only scale it. The output map remains linear because the mean of the positions is calculated from the output mean as $P^{-\top}\mu_y$.

A multilayer perceptron ϕ_{aug} with tanh activation functions for the hidden layers and a linear output layer is used for both $f^{\text{n}}_{\text{aug}}$ and $g^{\text{n}}_{\text{aug}}$:

$$\text{col}(f^{\text{n}}_{\text{aug}}, g^{\text{n}}_{\text{aug}}) = \phi_{\text{aug}}(\theta_{\text{aug}}, \hat{x}^{\text{n}}_k, u^{\text{n}}_k).$$

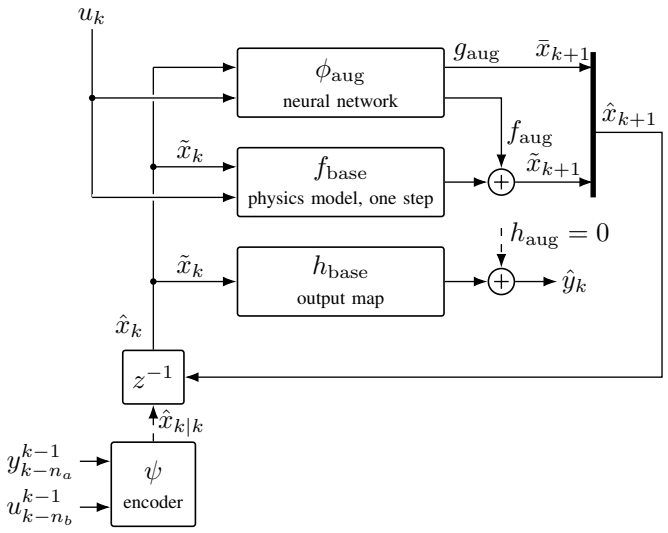


Fig. 3: Augmented model. The physics step f_{base} and the network ϕ_{aug} act in parallel on $\hat{x}_k$. ϕ_{aug} gives f_{aug} and g_{aug}, and the output map is not learned. The encoder ψ sets $\hat{x}$ at each window start. During training, the input is the closed-loop input of Section III-C. [**TODO:** No setpoint generator currently for the inputs; layout cramped at one column width.]

The inputs to the neural network are the $6 + n_{\bar{x}}$ states and the three actuator forces. The first six outputs $f_{\text{aug}}^{\text{n}}$ update the six physical states and the remaining $n_{\bar{x}}$ outputs $g_{\text{aug}}^{\text{n}}$ and equals the next additional states $\bar{x}_{k+1}$ (Fig. 3). These corrections act on all six physical states, the positions included, as in the dynamic-parallel structure [8, Table 1]. The positions are also corrected because within one sample period an unmodelled force would change both the velocity and position, correcting only for the velocities would not be able to capture this. Both f_{aug} and g_{aug} are given the payload position Y through $\tilde{x}_k$, enabling the neural network to learn model position-dependent effects.

Randomly initializing the learning components can give an unstable and bad optimization problem, to resolve this training starts from the baseline [8, Sec. 5.4]. At the start of training f_{aug} is set to zero, resulting in the augmented model being equal to the baseline. Because we make use of a feedforward neural network this only requires us to zero the output layer.

The gantry baseline model has two free axes, X and Y, without stiffness in (4), which are also affected by the learned correction. The discretized baseline model has a pole at $z = 1$ on each of these stiffness free axes, making it marginally stable. Therefore these axes act as integrators: a consant error in the correction keeps accumalating in the position. The alternative to only correct the velocities would not prevent the integrator issue, because an error in the velocity also accumulates in the position. The feedback controller is able to correct this drift, which is why we have chosen to simulate the model in closed-loop (Section III-C).

B. Encoder and initial state

- Each window starts from a state of which only the positions are measured; an encoder estimates it from the input-output history (Beintema); the window includes sample τ (encoder paper Eqs. (11) to (17))
- The encoder is a linear map plus a nonlinear correction (encoder paper Eq. (8)); the linear part acts on scaled, uncentred histories (D-055, code)
- W^b: linearise at rest at $Y = 0$, the design point of the controller; frozen at $Y = 0$ the baseline is LTI, so the linearisation is exact there (this work)
- W^b is the reconstructability map of the encoder paper, Eqs. (15) to (17); the noise-model version (Eqs. (18) to (29)) needs a process-noise gain, our noise is measurement noise
- Adaptations: marginally stable but observable baseline; co-estimated θ_{base} (footnote 1)
- Model-based rather than data-based (2026 Eq. (30)): the baseline simulation the latter needs drifts on X and Y; negligible cost (encoder paper Sec. 4.4) (this work)
- Only positions are measured, so an encoder estimates the state at each window start τ from past inputs and outputs
- Choice: short windows from encoded states
- Choice: the history includes sample τ; the encoder estimates all $6 + n_{\bar{x}}$ states:

Only the positions of the state at the start τ of a training window are measured. Following Beintema et al., an encoder estimates this state from the input and output history [20]. Simulating short windows from encoded states keeps the cost of a gradient step independent of the record length and stabilises the optimisation [20, Secs. 2, 3]. Unlike the criterion of Hoekstra et al., whose encoder reads the samples up to $\tau - 1$ [8, Eq. (22e)], the history includes sample τ, as the reconstructability map used below requires [21, Eqs. (11) to (17)]:

$$\begin{aligned} \hat{x}_{\tau|\tau}^{\text{n}} &= \psi(\theta_{\text{enc}}, \chi_\tau), \\ \chi_\tau &= \text{col}\big(y_\tau^{\text{n}}, \ldots, y_{\tau-n}^{\text{n}}, u_\tau^{\text{n}}, \ldots, u_{\tau-n}^{\text{n}}\big), \end{aligned} \tag{17}$$

- The encoder is a linear map plus a network correction; W^b gives the physical states, W^a the added states:

where n is the encoder history, $\chi_\tau \in \mathbb{R}^{6(n+1)}$ the recorded window and $\theta_{\text{enc}} \in \mathbb{R}^{n_{\text{enc}}}$ the encoder parameters. The encoder thus estimates all $6+n_{\bar{x}}$ states, the measured positions included. [**TODO:** Align the figure's past-only encoder-history labels with (17), which also uses the current sample.]

The encoder is a linear map with a nonlinear correction [21, Eq. (8)]. Its added-state rows W^a have no baseline counterpart. The linear part acts on the scaled but uncentred window, for which the map (**??**) below is built:

$$\psi(\theta_{\text{enc}}, \chi_\tau) = \begin{bmatrix} W^b \\ W^a \end{bmatrix} (\chi_\tau + \chi_0) - \begin{bmatrix} T_x \mu_x \\ 0 \end{bmatrix} + \tilde{\psi}(\chi_\tau), \tag{18}$$

- Choice: W^b from the baseline, linearised at rest at $Y = 0$ with nominal parameters, ZOH over T_s:

where $W^b \in \mathbb{R}^{6\times 6(n+1)}$, $W^a \in \mathbb{R}^{n_{\bar{x}}\times 6(n+1)}$, χ_0 restores the means that (15) removes, $\tilde{\psi}$ is a neural network, and θ_{enc} collects W^b, W^a and the parameters of $\tilde{\psi}$. We write ψ_a for the last $n_{\bar{x}}$ rows of ψ and θ^a_{enc} for W^a with the added-state rows of the output layer of $\tilde{\psi}$. [**TODO:** VERIFY bib: `hoekstra2026encoder` = arXiv:2602.13108v1 (13 Feb 2026), Hoekstra, Györök, Tóth, Schoukens; matches PDF p. 1. Check for a published (IFAC) version before submission.]

The physical rows start from the baseline, through the model-based initialisation of Hoekstra et al. [21, Sec. 3.1]. Its map needs a linear model, so as in [21, Eqs. (30) to (32)] we linearise the baseline at an equilibrium. We take rest at $Y = 0$, which is an equilibrium for zero input because K in (4) vanishes on X and Y, and the position at which the controller of the records is designed (Section V). Frozen at $Y = 0$, (2) is linear, so the linearisation is the baseline itself at that position and approximates it only as Y moves. Discretising by zero-order hold over T_s and scaling as (15) without the means gives

$$A_d = T_x e^{A_c T_s} T_x^{-1}, \quad B_d = T_x \int_0^{T_s} e^{A_c s}\,\mathrm{d}s\, B_c T_u^{-1}, \quad C_d = T_y C_c T_x^{-1},$$
$$A_c = \begin{bmatrix} 0 & I_3 \\ -M_0^{-1}K & -M_0^{-1}C \end{bmatrix}, \quad B_c = \begin{bmatrix} 0 \\ M_0^{-1}P \end{bmatrix}, \quad C_c = \begin{bmatrix} P^\top & 0 \end{bmatrix}, \tag{19}$$

with $M_0 = M(0)$ from (5), evaluated at the nominal parameters. Stacking the outputs and inputs of the window as Y_τ and U_τ, the output and input parts of $\chi_\tau + \chi_0$ with the newest sample first, gives $Y_\tau = O_n T_x \tilde{x}_{\tau-n} + \Gamma_n U_\tau$ and $T_x\tilde{x}_\tau = A_d^n T_x \tilde{x}_{\tau-n} + r_n U_\tau$. Eliminating $T_x\tilde{x}_{\tau-n}$ gives the reconstructability map [21, Eqs. (11) to (17)]

- The baseline is marginally stable on X and Y but observable from the positions, so O_n has full column rank
- Choice: the noise-free map of Hoekstra et al., not the noise-model version, since no process-noise gain is known
- W^b only sets the start; it is trained with all other parameters, including θ_{base}, which Hoekstra et al. keep fixed

where $O_n = \operatorname{col}(C_d A_d^n, \ldots, C_d A_d, C_d) \in \mathbb{R}^{3(n+1)\times 6}$, Γ_n is block upper triangular with block (i,j) equal to $C_d A_d^{j-i-1} B_d$ for $j > i$ and zero otherwise, $r_n = [0 \;\; B_d \;\; A_d B_d \;\; \cdots \;\; A_d^{n-1} B_d]$, as in [21, Eqs. (10), (13)], and $^+$ is the Moore-Penrose pseudoinverse. The paper assumes a stable baseline [21, Sec. 2]. Ours is marginally stable, but the measured positions make (A_d, C_d) observable, so O_n has full column rank. The map is derived without noise; with white Gaussian measurement noise it is the maximum-likelihood estimate of the state [21, Sec. 3.1]. The paper also gives a version for process noise, which needs a known noise gain [21, Eqs. (18) to (29)]. The noise in our records enters at the position measurement (Section V) and no noise gain is known, so we use (**??**). Our measurement noise is coloured, so the map is a starting value for W^b, not an optimal estimator. It leaves co-estimation of the baseline parameters out of scope [21, footnote 1], whereas here W^b is trained jointly with them. [**TODO:** Fix before the final run (`FINAL-RUN-FIXES.md` item 1): W^b is built from the nominal parameters, which equal those of the simulated truth, while θ_{base} starts detuned. The final runs build W^b at the start $\theta_{\text{base},0}$; the text then states that.]

Hoekstra et al. instead fit the baseline encoder to states simulated with the baseline [8, Eq. (30)], the data-based alternative of [21, Sec. 3.2]. On the gantry, that simulation is driven by the recorded input and drifts on the free axes X and Y (Section III-C), so it would fit W^b to wrong states. [**TODO:** Revise: the drift holds only for the open-loop simulation of Jan's Eq. (30). A closed-loop simulation of the baseline with the known controller does not drift (Section III-C), so drift alone is no reason against the data-based route. Candidate: the closed-loop alternative would need an extra simulation, which the model-based map avoids.] The map (**??**) needs no simulation and negligible computation [21, Sec. 4.4]. It gives better starting values and faster convergence than a random encoder, at comparable final accuracy [21, Sec. 4.4].

C. Closed-loop simulation

- The model is not driven by the recorded input (2026 Eq. (22)), for two reasons: noise-induced bias under feedback (Kessels Remark 5.6, SUBNET; indirect approach, Forssell and Ljung) and the free axes of Section III-A, which violate the stability condition behind SUBNET consistency (Beintema Cond. 1); motivation, not a proof
- As in Kessels (Eqs. (5.12), (5.13)), each window is simulated in closed loop with the known controller, which forms its error against the model output; sytem and model loops share r and u^{ff}
- The controller is linear and time invariant, so subtracting the sytem's loop cancels r and u^{ff}; the residual form needs only the recorded u_{act} and y, and is exact when u_{act} was produced by this controller from y (this work)
- $x^c_\tau = 0$ starts the model's controller in the sytem's controller state, as the encoder does for the plant state; no controller state is estimated, unlike Kessels Remark 5.4 (this work)
- No plant feedthrough fixes the step order, so D_{fb} closes no algebraic loop (as Kessels Remark 5.1); one normalised step displayed
- The output error re-enters the model input, so part of it is corrected within the window; error before τ is not carried in, the free-run validation carries it

The data is measured in closed-loop, and the model is simulated in closed-loop. Instead of using the recorded input as in [8, Eq. (22)], we include the feedback controller in the simulation for the following reasons. The feedback controller causes a correlation between the recorded input and the measurement noise. Kessels was not able to identify an open-loop SUBNET model using closed-loop data from a wire bonder. He suggest that this could be explained by noise-induced bias due to this correlation and the integral term of the controller [13, Remark 5.6].

The indirect approach for linear systems with a known controller fits the closed-loop model using the reference.

Because the reference is uncorrelated with the noise, the model can be identified from the output prediction error [22]. [**TODO:** output prediction error??]

Also, as has been established in Section III-A: the free axes X and Y make the open-loop gantry marginally stable. SUBNET assumes that the system that generates the data is incrementally exponentially output stable [20, Cond. 1]. The identification method we have adapted from [8, Sec. 5.5] relies on this SUBNET assumption. The open-loop gantry system does not satisfy this assumption: a mismatch in the initial velocity will create a permanent position offset. These reasons motivate the choice for simulating in closed-loop. They do not prove that our estimate is consistent. [**TODO:** ?] [**TODO:** Missing: section number for [22]. The indirect approach was verified in Sec. 5.2.1 of the LiU report LiTH-ISY-R-2021; the Automatica numbering is unchecked. Candidate: cite without a section number until the journal version is read.] [**TODO:** Missing: D-140 (Ruled out (4)) bars Kessels' Remark 5.6 as an argument because the records were then noise free. The thesis records are noisy, so the ban appears lapsed. Candidate: keep the sentence and amend D-140 (basis: noisy thesis records, Section V).] [**TODO:** Missing: evidence on the thesis data that open-loop rollouts accumulate position error on X and Y. Candidate: the open-loop versus closed-loop drift on one record that `THESIS-RESULTS.md` Sec. 5 plans, reported in Section VI.]

Every training window is simulated in closed-loop with the known controller, as in Kessels [13, Eqs. (5.12), (5.13)]. The controller is linear and time invariant, the model output is used to calculate the position error in actuator coordinates [**TODO:** qact?]. The controller calculates the actuator forces from this error. The system and model use the same reference r and feedforward u^{ff}, but use their own output for feedback, with controller states x^{fb} and $\hat{x}^{\mathrm{fb}}$ (Fig. 4):

$$\begin{aligned} x^{\mathrm{fb}}_{k+1} &= A_{\mathrm{fb}} x^{\mathrm{fb}}_k + B_{\mathrm{fb}}(r_k - y_k), \\ u_{\mathrm{act},k} &= C_{\mathrm{fb}} x^{\mathrm{fb}}_k + D_{\mathrm{fb}}(r_k - y_k) + u^{\mathrm{ff}}_k, \\ \hat{x}^{\mathrm{fb}}_{k+1} &= A_{\mathrm{fb}} \hat{x}^{\mathrm{fb}}_k + B_{\mathrm{fb}}(r_k - \hat{y}_k), \\ \hat{u}_{\mathrm{act},k} &= C_{\mathrm{fb}} \hat{x}^{\mathrm{fb}}_k + D_{\mathrm{fb}}(r_k - \hat{y}_k) + u^{\mathrm{ff}}_k, \end{aligned} \tag{20}$$

where $(A_{\mathrm{fb}}, B_{\mathrm{fb}}, C_{\mathrm{fb}}, D_{\mathrm{fb}})$ are the state-space matrices of the controller discretised with sampling time T_s. Both r_k and u^{ff}_k are in actuator coordinates. The model uses the calculated actuator force $\hat{u}_{\mathrm{act},k}$ instead of the recorded $u_{\mathrm{act},k}$ in (13).

Since we use a linear controller we are able to substract the system and model controller equations to cancel r and u^{ff}. This leaves us with the controller state difference $x^c = \hat{x}^{\mathrm{fb}} - x^{\mathrm{fb}}$, driven by $y - \hat{y}$. The model is still simulated with respect to the reference r, but r and u^{ff} are not used in simulation. If the controllers between the system and simulation differ, the equations are only approximate and noise-induced bias cannot be ruled out.

At the start of each training window we set $x^c_\tau = 0$, which sets the model's controller state equal to that of the sytem at τ. Therefore the encoder only has to initialize the states of the model. Kessels instead uses the system's controller state at the start of each window, either measured directly or calculated by simulating the controller from the start of the record [13, Remark 5.4]. We then write the closed-loop model in (20) as

$$\begin{aligned} x^c_{k+1} &= A_{\mathrm{fb}} x^c_k + B_{\mathrm{fb}}(y_k - \hat{y}_k), \qquad x^c_\tau = 0, \\ \hat{u}_{\mathrm{act},k} &= u_{\mathrm{act},k} + C_{\mathrm{fb}} x^c_k + D_{\mathrm{fb}}(y_k - \hat{y}_k), \end{aligned} \tag{21}$$

which needs only the recorded u_{act} and y. Within a window, $\hat{y}$ is still the response of the model loop to r and u^{ff} from the states at τ, although neither signal is used. [**TODO:** Check for Setup: the training loop uses the controller re-discretised at the model rate, while the records were generated at a higher rate. Run the re-discretised controller on the recorded $r - y$ and compare its feedback force with the recorded one. A rollout with $\hat{y} = y$ gives $\hat{u} = u$ by construction and checks nothing. Related: `FINAL-RUN-FIXES.md` item 3, the D-232 leak check on the lowpass data, passed (0.22 to 0.50 nm).]

The output of the augmented model depends on its state only, so each sample is evaluated in a fixed order: the output, then the error and the model input, then the model and controller states advance. The controller feedthrough D_{fb} therefore closes no algebraic loop, for the same reason that Kessels' output augmentation does not depend on the input [13, Remark 5.1]. In the coordinates of (15), where the output means cancel in the error, one step of a window started at τ is

$$\begin{aligned} \hat{y}^{\mathrm{n}}_{k|\tau} &= C^{\mathrm{n}} \tilde{x}^{\mathrm{n}}_{k|\tau}, \qquad \varepsilon_{k|\tau} = T_y^{-1}\big(y^{\mathrm{n}}_k - \hat{y}^{\mathrm{n}}_{k|\tau}\big), \\ \hat{u}^{\mathrm{n}}_{k|\tau} &= u^{\mathrm{n}}_k + T_u\big(C_{\mathrm{fb}} x^c_k + D_{\mathrm{fb}} \varepsilon_{k|\tau}\big), \\ x^c_{k+1} &= A_{\mathrm{fb}} x^c_k + B_{\mathrm{fb}} \varepsilon_{k|\tau}, \qquad x^c_\tau = 0, \end{aligned} \tag{22}$$

after which $\hat{x}^{\mathrm{n}}_{k+1|\tau}$ follows from (13) with input $\hat{u}^{\mathrm{n}}_{k|\tau}$, starting from the encoder state (17). Here $\varepsilon_{k|\tau} = y_k - \hat{y}_{k|\tau}$ is the output error in metres.

The output error of the model thus re-enters its own input through the controller, so part of a model error is corrected within the window, as on the sytem. Model error made before τ is not carried into the window, because the model state and x^c both restart at τ. The free-run validation of Section III-D applies (22) over whole records and does carry it.

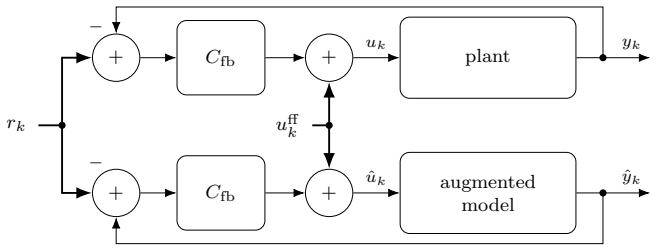


Fig. 4: Recorded plant loop (top) and augmented-model loop (bottom), driven by one shared reference r_k and feedforward u^{ff}_k, with the same feedback controller. The model output is evaluated before the input moves its state to the next sample.

D. Identification problem

- θ_{base} is estimated jointly with network and encoder, as in 2026 Eq. (22); Kessels fixes the first-principles parameters (Eq. (5.12))

- Training coordinates ξ, zero at the start; nine positive combinations logarithmic, the signed m_Δ through (23), so every iterate satisfies (12) without clamp or penalty (D-229)
- One display: cost over $\vartheta = (\xi, \theta_{\text{aug}}, \theta_{\text{enc}})$ with the model, encoder and loop constraints; every window sample scored; controller fixed; a closed-loop fit can hide plant error (Section VI)
- Adam with exact backpropagation through the window, then an unbounded L-BFGS polish (ξ admissible for every value); best validation free-run checkpoint; polish kept only if it improves validation and no meter regresses
- Initialisation display: $\xi = 0$, zero output layer, encoder start; reproduces the baseline (2026 Eq. (29)); $\mathcal{M}_{\text{U}}$ named
- Choice: θ_{base} estimated together with network and encoder, in coordinates ξ that are zero at the start
- Nine combinations logarithmic, m_Δ bounded:

We estimate θ_{base} jointly with the network and the encoder, as Hoekstra et al. do [8, Eq. (22)], whereas Kessels leaves the first-principles parameters fixed [13, Eq. (5.12)]. Every iterate must satisfy (12), so that $M(Y) \succ 0$ and the LFR stays well posed without a clamp or a penalty. We therefore train in coordinates $\xi \in \mathbb{R}^{10}$ that are zero at the start $\theta_{\text{base},0}$. The nine positive combinations are logarithmic, $\theta_{\text{base},j} = \theta_{\text{base},0,j}\, e^{\xi_j}$, which keeps them positive. The signed m_Δ is bounded by (12) itself:

$$m_\Delta = \rho \, \frac{2}{L_b} \sqrt{m_\Sigma J_{\text{eff}}} \, \tanh(\eta_0 + s\, \xi_\Delta), \tag{23}$$

- Every sample of each window scored, controller fixed:

where ξ_Δ is the component of ξ for m_Δ, $\rho \in (0, 1)$ a margin, and η_0 and s reproduce $m_{\Delta,0}$ and its sensitivity at $\xi = 0$ (Appendix A-D). Every ξ thus gives $M(Y) \succ 0$ for all Y, which guarantees positive definite inertia but not positive raw masses.

Over the window starts $\mathcal{T}$ and the window length n_f, the trained parameters $\vartheta = (\xi, \theta_{\text{aug}}, \theta_{\text{enc}})$ minimise the normalised closed-loop output error, with the controller fixed. As in [8, Eq. (22)], every sample of a window is scored. We adapt that criterion in two respects: the model is simulated in the loop of Section III-C, and the encoder reads sample τ. Because the controller acts on the output error, a good closed-loop fit can mask plant error, so Section VI also tests the model under a changed controller. The identification problem is

$$\begin{aligned}
&\min_{\vartheta} \; V(\vartheta) = \frac{1}{|\mathcal{T}|\, n_f\, n_y} \sum_{\tau \in \mathcal{T}} \sum_{k=\tau}^{\tau+n_f-1} \left\| y^{\text{n}}_k - \hat{y}^{\text{n}}_{k|\tau} \right\|_2^2 \quad \text{s.t.} \\
&\tilde{x}^{\text{n}}_{k+1|\tau} = f^{\text{n}}_{\text{base}}\big(\theta_{\text{base}}(\xi), \tilde{x}^{\text{n}}_{k|\tau}, \hat{u}^{\text{n}}_{k|\tau}\big) + f^{\text{n}}_{\text{aug}}\big(\theta_{\text{aug}}, \hat{x}^{\text{n}}_{k|\tau}, \hat{u}^{\text{n}}_{k|\tau}\big), \\
&\bar{x}_{k+1|\tau} = g^{\text{n}}_{\text{aug}}\big(\theta_{\text{aug}}, \hat{x}^{\text{n}}_{k|\tau}, \hat{u}^{\text{n}}_{k|\tau}\big), \\
&\hat{x}^{\text{n}}_{\tau|\tau} = \psi(\theta_{\text{enc}}, \chi_\tau), \quad \hat{y}^{\text{n}}_{k|\tau},\, \hat{u}^{\text{n}}_{k|\tau} \text{ from (22)},
\end{aligned} \tag{24}$$

- Choice: Adam over the whole window, then L-BFGS polish; best validation free-run checkpoint kept
- Training starts from the baseline:

for $\tau \le k < \tau + n_f$, where $n_y = 3$ and $\theta_{\text{base}}(\xi)$ is the coordinate map above. The controller matrices are fixed, but they are part of the computation: since $\hat{u}^{\text{n}}_{k|\tau}$ depends on $\hat{y}^{\text{n}}_{k|\tau}$, the gradient of V also passes through the controller recursion.

Because f_{aug} acts on the same rows as f_{base}, it can also learn changes that a different θ_{base} would produce, so the split between baseline and network is not unique [8, Sec. 5.2]. With a non-unique θ_{base}, the baseline can lose its physical meaning [11, Sec. 3]. Here, the estimated combinations are interpretable if they keep that meaning, which requires a unique estimate. Section IV extends the method so that this estimate is unique.

We minimise V with Adam on minibatches of windows, with the gradient obtained by automatic differentiation through the whole window, so no part of the window is truncated from the gradient. We then polish with L-BFGS on a fixed set of windows. Drenth et al. also follow Adam with L-BFGS, in its bounded form L-BFGS-B [16, Sec. 5]. Our polish needs no bounds, because (23) and the logarithmic coordinates give an admissible baseline for every ξ. During Adam, the parameters with the lowest validation error are kept, the closed-loop free-run error over whole validation records of Section V, each simulated from one encoder start with $x^c = 0$. The polish is kept only if it lowers this error and no monitored quantity regresses. The encoder history n, the window length n_f, the added-state dimension $n_{\bar{x}}$, the width and depth of ϕ_{aug} and the optimiser settings are given with their basis in Section V. [**TODO:** Fix before the final run (`FINAL-RUN-FIXES.md` item 2): the polish acceptance meters include `combo_err`, the error of the ten combinations against the true parameters, and possibly `param_loss`, the drift from the start. The final runs accept on the validation error and non-oracle meters only.]

Training starts from the baseline, as Hoekstra et al. require of the initialisation [8, Eq. (29)]. Every trained block starts as

$$\begin{aligned}
&\xi = 0 \implies \theta_{\text{base}} = \theta_{\text{base},0}, \\
&W_L = 0,\; b_L = 0 \implies f^{\text{n}}_{\text{aug}} = 0,\; g^{\text{n}}_{\text{aug}} = 0, \\
&W^b \text{ from the reconstructability map}, \quad W^a \sim \mathcal{U}\big(-a, a\big), \quad \tilde{W}_L = 0,\; \tilde{b}_L =
\end{aligned} \tag{25}$$

- The model reproduces the baseline at the start; this model is called $\mathcal{M}_{\text{U}}$

where $\theta_{\text{base},0}$ are the starting combinations (Section V), $W_L \in \mathbb{R}^{(6+n_{\bar{x}}) \times n_h}$ and $b_L \in \mathbb{R}^{6+n_{\bar{x}}}$ the weight and bias of the output layer of ϕ_{aug} with n_h its hidden width, $\tilde{W}_L$ and $\tilde{b}_L$ those of $\tilde{\psi}$, and $a = \sqrt{6/(n_{\bar{x}} + 6(n+1))}$ the Xavier bound, as in [8, Eq. (31)]. The hidden layers of ϕ_{aug} and $\tilde{\psi}$ keep the default random initialisation of the software. From the same physical initial state, the augmented model then reproduces the baseline. [**TODO:** Attribution: the all-zero output layer follows the authors' public dynamic-parallel code (D-237) and is the thesis choice (D-240); [8, Sec. 5.4.3] instead starts the learning component as a linear map with random entries. Ask Jan which produced the reported S-DP results.]

We call $\mathcal{M}_{\mathrm{U}}$ the augmented model (13) identified by (24) from the start (25). Its input is the actuator force u_{act} and its output the actuator positions q_{act}.

E. Added-state initialisation

- Problem: the start (25) gives $\tilde{x}$ a model (f_{base}, W^b, output) and $\bar{x}$ none (random W^a, random added-state rows of ϕ_{aug}); V scores only the output and f_{aug} can lower the same error statically, so training can leave the omitted dynamics out of $\bar{x}$ (Venkatraman Sec. 2.1; Section VI tests it)
- PS is this work's, adapted from two-stage regression (Hefny Sec. 2) and its use before BPTT (Downey Sec. 4.2); it gives $\bar{x}$ a start from data, as W^b gives $\tilde{x}$ one from the baseline
- S1: a decoder on ψ_a predicts the current model's residual over $\tau+1$ to $\tau+M$; the minimiser is the conditional mean, so $\bar{x}_{\tau|\tau}$ becomes a predictive state; trains θ_{p}, θ^a_{enc} only, one update after every update of V
- Stopping rule: ν on calibration windows ends S1 (plateau or cap) and decides whether S2 runs
- S2: once, one-step regression of $g^{\mathrm{n}}_{\mathrm{aug}}$ on frozen encoder pairs with the recorded input, which equals the loop input when the encoder reproduces y_{k_i}; overlapping windows (?)
- Hand-over: decoder discarded, (24) continues from ϑ_{PS}; deployed model, criterion and selection unchanged; $\bar{x}$ latent; $\mathcal{M}_{\mathrm{PS}}$ named
- The current runs differ in two points: g_{aug} starts at zero, and the S1 target starts at τ (?)
- The start (25) gives the physical states a model and the added states none
- V scores only the output, and f_{aug} can lower the error without memory, so training can leave the omitted dynamics out of $\bar{x}$

The physical states of the baseline are initialized with the reconstructability map W^b (25) and updated through $f^{\mathrm{n}}_{\mathrm{base}}$, while the added-state rows W^a are randomly initialized. For the physical states their generalized coordinates q_k are directly mapped to actuator positions, $\hat{y}_k = P^\top q_k$ in (13). The added states are updated through the random added-state rows of ϕ_{aug}, without their initialization or initial updates being derived from the baseline or the data. The output $\hat{y}_k = P^\top q_k$ contains no added state, and $\bar{x}_k$ enters $\tilde{x}_{k+1}$ only through $f^{\mathrm{n}}_{\mathrm{aug}}$ in (24). This means that for the added states V only updates their effect on the output through $f^{\mathrm{n}}_{\mathrm{aug}}$. V contains nothing that directly incentivises $\bar{x}$ to predict the resulting output error [?, Sec. 2.1]. This gives $f^{\mathrm{n}}_{\mathrm{aug}}$ the capabilities to lower part of this error with $\tilde{x}$ and $\hat{u}$ alone. Minimising V can therefore reduce the error without capturing the missing dynamics in $\bar{x}$. [**TODO:** Section VI tests whether it does.] This forms a sharp contrast between the physical and added learning components, due to the physical states being able to directly map to the output, while also having a much better starting point.

[**TODO:** Current runs: the 2026-10-11 runs (R1 to R7) start the whole output layer of ϕ_{aug} at zero (`G_INIT` zero, D-240), as the authors' public code does (D-237), so $g^{\mathrm{n}}_{\mathrm{aug}} = 0$ and $\bar{x}$ is zero after the first step. The final runs keep the added-state rows random, as [8, Eq. (29), Sec. 5.4.3] prescribes (D-248, `FINAL-RUN-FIXES.md` item 6); (25) in Section III-D then changes accordingly. PS with a live start is so far one seed with the ResNet variant (4.0×10^{-6} m against 1.0×10^{-5} m for plain training).]

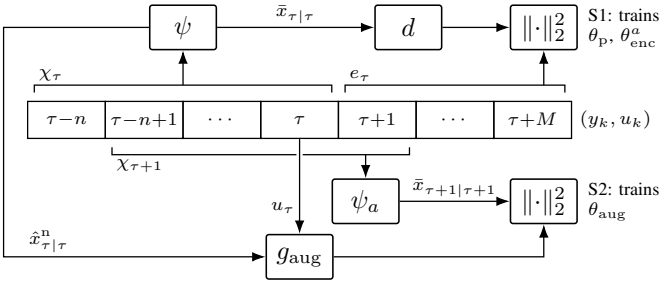


Fig. 5: Windows of PS on one record, layout adapted from [23, Fig. 2]. S1 (top): the encoder maps the window χ_τ to $\hat{x}^{\mathrm{n}}_{\tau|\tau}$, and the decoder d maps its added part $\bar{x}_{\tau|\tau}$ to the residuals e_τ of the next M samples (26). S2 (bottom): g_{aug} maps $\hat{x}^{\mathrm{n}}_{\tau|\tau}$ and the recorded u_τ to the added state encoded from the window shifted by one sample (28). [**TODO:** The figure shows the target from $\tau+1$ (`FINAL-RUN-FIXES.md` item 5). The 2026-10-11 runs start it at τ, so the noise in y_τ enters both χ_τ and e_τ.]

- Choice: predictive-state (PS) initialisation gives the added states a role from data, as W^b does for the physical states, then hands over to (24)
- Adapted from two-stage regression and its use before backpropagation through time; differences: residual target, nonlinear transition, first stage alongside V

We therefore propose predictive-state (PS) initialisation, which is a data-driven initialisation for the added states. Similar to W^b after its initialisation is done, the states optimize through (24). PS is a two-stage regression of predictive-state learning [23, Sec. 2], and it makes use of initialisation before backpropagation through time [24, Sec. 4.2].

The first stage (S1) predicts the residual of the model, instead of features of future observations as in [23]. The second stage (S2) trains the nonlinear state update instead of a linear operator. We also use the first stage during the initial training of V, instead of before training as in [24]. Fig. 5 shows which samples are used for both stages.

- S1, during the first updates of (24): a temporary decoder on $\bar{x}_{\tau|\tau}$ predicts the current model's residual over the M samples after the encoder window:

Stage S1 trains the encoder's added-state outputs to predict the residual of the augmented model (13) using the model parameters ϑ from the latest V update step. Each update of V is followed by one S1 update with its own Adam optimiser. For the S1 update we select a minibatch $\mathcal{B}$ of samples τ from the training data. The training data is divided into excitation classes ([**TODO:** see section]), and for classes with at least

three records one is left out of S1 and S2 to evaluate ν (27). The left out data is still used in V. For each sample τ, the encoder uses the last $n+1$ samples up to τ, χ_τ, to estimate the initial state $\hat{x}^{\mathrm{n}}_{\tau|\tau}$. From this state the augmented model is simulated in the closed loop of Section III-C. We calculate the residual from the next $M = n+1$ samples, from $\tau+1$ to $\tau+M$. The top of Fig. 5 shows the window χ_τ and the samples of the residual e_τ. By convention we have chosen to use the same number of samples as the encoder window. Then a temporary decoder d with only the added states as inputs $\bar{x}_{\tau|\tau} = \psi_a(\theta_{\mathrm{enc}}, \chi_\tau)$ is used to predict this residual:

$$
\begin{aligned}
e_\tau &= \frac{1}{\sigma_e}\,\mathrm{col}\big(y^{\mathrm{n}}_k - \hat{y}^{\mathrm{n}}_{k|\tau}\big)_{k=\tau+1}^{\tau+M},\\
d(\theta_{\mathrm{p}}, \bar{x}) &= W_{\mathrm{p},2}\tanh\big(W_{\mathrm{p},1}\bar{x} + b_{\mathrm{p},1}\big) + b_{\mathrm{p},2},\\
&\min_{\theta_{\mathrm{p}},\,\theta^a_{\mathrm{enc}}} \frac{1}{|\mathcal{B}|\,M\,n_y}\sum_{\tau\in\mathcal{B}}\big\|d\big(\theta_{\mathrm{p}}, \psi_a(\theta_{\mathrm{enc}}, \chi_\tau)\big) - e_\tau\big\|_2^2,
\end{aligned}
\tag{26}
$$

- The minimiser is the conditional mean of the residual given the window; through the bottleneck $\bar{x}_{\tau|\tau}$ the added states become a predictive state
- The target starts after the window, so the two share no sample

where $\hat{y}^{\mathrm{n}}_{k|\tau}$ is the output from the closed-loop simulation of Section III-C, starting at $\hat{x}^{\mathrm{n}}_{\tau|\tau}$. This simulated output is kept fixed during the S1 update. $\mathcal{B}$ is the minibatch of samples τ and σ_e is the RMS of all residual values in $\mathcal{B}$. The decoder $d : \mathbb{R}^{n_{\bar{x}}} \to \mathbb{R}^{Mn_y}$ has parameters $\theta_{\mathrm{p}} = (W_{\mathrm{p},1}, b_{\mathrm{p},1}, W_{\mathrm{p},2}, b_{\mathrm{p},2})$, with $W_{\mathrm{p},1} \in \mathbb{R}^{n_h \times n_{\bar{x}}}$ and $W_{\mathrm{p},2} \in \mathbb{R}^{Mn_y \times n_h}$. Both $W_{\mathrm{p},2}$ and $b_{\mathrm{p},2}$ are initially set to zero, so that the decoder starts by predicting zero residual. Only θ_{p} and θ^a_{enc} are updated by S1. The parameters of ϑ, including θ^a_{enc} are still trained through V.

The conditional mean $\mathbb{E}[e_\tau \mid \chi_\tau]$ is used to minimise the expected squared error. The decoder only sees up to $\bar{x}_{\tau|\tau}$, not the full history χ_τ, this forces the encoder to use the added states to predict the future residual. [**TODO:** Instead of using features of future observations as Hefny et al. [23, Sec. 2] for predictive-state learning, we use the model residual.] To avoid correlation between the history and future noise Hefny et al. separates the past and future samples [23, Sec. 2]. We borrow this principle by having the residual starts at $\tau+1$, so that there is no overlap between the samples used for the encoder and to calculate the residual. [**TODO:** Venkatraman et al. use a similar approach by predicting future observations from the internal states during training [?, Sec. 3].] For the PS implementation the decoder predicts the model residual and is removed after S1. [**TODO:** Current runs (`FINAL-RUN-FIXES.md` item 5): the 2026-10-11 runs start the target at τ (`ps2_opening.py`, `_sample` and `_residual`), so sample τ is both the last sample of χ_τ and the first of e_τ, and the noise in y_τ enters both sides of (26).] [**TODO:** Missing: Hefny et al. model the noise as independent per sample (Fig. 1). Our measurement noise is low-pass and fed back by the controller, so disjoint windows remove the shared sample but not all correlation between χ_τ and the noise in e_τ. Candidate: one sentence here or in the Discussion (basis: D-238 noise model).]

- Unexplained fraction ν on fixed calibration windows:

To evaluate how well $\bar{x}_{\tau|\tau}$ performs at predicting the residual, a fixed set $\mathcal{C}$ of windows from the held out data is used. For this set the squared prediction error of the decoder divided by the squared residual is computed,

$$
\nu = \frac{\sum_{\tau\in\mathcal{C}}\big\|d\big(\theta_{\mathrm{p}}, \psi_a(\theta_{\mathrm{enc}}, \chi_\tau)\big) - e_\tau\big\|_2^2}{\sum_{\tau\in\mathcal{C}}\|e_\tau\|_2^2}. \tag{27}
$$

- S1 ends when ν stops improving or at a cap; S2 follows only if ν is below a threshold, otherwise the S1 changes are kept (exact rule in the appendix)

S1 is stopped when a maximum number of updates is reached or when ν stops improving. S2 starts only if ν is below a threshold $\nu_{\max}$, else the decoder is removed and training continues. In both cases the encoder changes from S1 are kept. The threshold rule for S2 is given in Appendix A-E. [**TODO:** Missing: basis of the PS switch and stop rules (S1 plateau and cap, screen update and threshold, S2 plateau and cap, snapshot sizes, calibration split as the last record of each class with three or more records). Candidate: rows of the hyperparameter table in Section V, all HEURISTIC (`PS2Spec`; `PS2-METHOD.md` Sec. 5), noting that the S1 cap, not the plateau, ended S1 in every run so far.]

- S2, once at the end of S1: one-step regression of $g^{\mathrm{n}}_{\mathrm{aug}}$ on frozen encoder pairs with the recorded input:

S1 has only trained the part of the encoder that initializes the added states $\bar{x}_{\tau|\tau}$, $g^{\mathrm{n}}_{\mathrm{aug}}$ is only updated by V. S2 trains the states itself. In S2 $g^{\mathrm{n}}_{\mathrm{aug}}$ is trained to predict the next added state $\bar{x}_{k+1}$ from the full state $\hat{x}^{\mathrm{n}}_k$ and the input u^{n}_k. For S2 samples k_i from the training data are selected, except the same held of records from S1 are left out. For each sample the encoder estimates $\hat{x}^{\mathrm{n}}_{k_i|k_i}$ from χ_{k_i} and $\bar{x}_{k_i+1|k_i+1}$ from the next window χ_{k_i+1}. Fig. 5 shows an example with $k_i = \tau$. The window χ_τ at the top is used for $\hat{x}^{\mathrm{n}}_{k_i|k_i}$, and the shifted window at the bottom is used for $\bar{x}_{k_i+1|k_i+1}$. We calculate these pairs once after S1, this set $\mathcal{P}$ is kept fixed.

During S2 the model is not simulated in closed-loop and V is paused. We instead train $g^{\mathrm{n}}_{\mathrm{aug}}$ on $\mathcal{P}$ using one-step regression with the recorded actuator force $u^{\mathrm{n}}_{k_i}$ as input, and a separate Adam optimiser:

$$
\begin{aligned}
\mathcal{P} &= \big\{(\hat{x}^{\mathrm{n}}_{k_i|k_i},\, u^{\mathrm{n}}_{k_i},\, \bar{x}_{k_i+1|k_i+1})\big\}_{i=1}^{N_{\mathcal{P}}},\\
\hat{x}^{\mathrm{n}}_{k_i|k_i} &= \psi(\theta_{\mathrm{enc}}, \chi_{k_i}), \quad \bar{x}_{k_i+1|k_i+1} = \psi_a(\theta_{\mathrm{enc}}, \chi_{k_i+1}),\\
&\min_{\theta_{\mathrm{aug}}} \frac{\sum_{i\in\mathcal{B}}\Big\|g^{\mathrm{n}}_{\mathrm{aug}}\big(\theta_{\mathrm{aug}}, \hat{x}^{\mathrm{n}}_{k_i|k_i}, u^{\mathrm{n}}_{k_i}\big) - \bar{x}_{k_i+1|k_i+1}\Big\|_2^2}{\sum_{i\in\mathcal{B}}\sum_{j=1}^{n_{\bar{x}}}\big(\bar{x}_{j,k_i+1|k_i+1} - m_j\big)^2},
\end{aligned}
\tag{28}
$$

- The recorded input is the one that moved the system from k_i to k_i+1; at a window start the loop input is:

where θ_{enc} is not updated during S2, $\mathcal{B} \subset \mathcal{P}$ is a minibatch, $\bar{x}_{j,\cdot}$ is the jth added state and m_j is the mean of $\bar{x}_{j,k_i+1|k_i+1}$

over $\mathcal{B}$. The cost is the fraction of the variance in the next added states that is not explained by the prediction.
The parameters with the lowest cost on the separate set of pairs from the held-out records are kept. For S2 we use the recorded actuator force $u^{\mathrm{n}}_{k_i}$ applied to the system at k_i. The encoder estimates $\bar{x}_{k_i+1|k_i+1}$ from χ_{k_i+1}, using the data up to k_i+1. At the start of a window in (21) $x^c = 0$, resulting in the model input

$$\hat{u}_{\mathrm{act},k_i} = u_{\mathrm{act},k_i} + D_{\mathrm{fb}}\big(y_{k_i} - \hat{y}_{k_i|k_i}\big),$$

- So it equals the recorded input when the encoder reproduces y; S2 leaves the physical rows of the output layer unchanged

when the encoder reproduces y_{k_i} this equals the recorded input, because at this sample only the controller feedthrough D_{fb} acts on the output error. For S2 $g^{\mathrm{n}}_{\mathrm{aug}}$ is therefore trained with approximately the input it receives at that sample in the closed-loop simulation of V. S2 updates all parameters θ_{aug} of ϕ_{aug}. The cost only depends on the added-state rows, so the physical rows of W_L are not changed. The hidden layers of ϕ_{aug} are shared by $f^{\mathrm{n}}_{\mathrm{aug}}$ and $g^{\mathrm{n}}_{\mathrm{aug}}$, so S2 also updates $f^{\mathrm{n}}_{\mathrm{aug}}$. [**TODO:** Missing: χ_{k_i} and χ_{k_i+1} share n samples, so the measurement noise in these samples enters both $\hat{x}^{\mathrm{n}}_{k_i|k_i}$ and $\bar{x}_{k_i+1|k_i+1}$. Regressing samples of overlapping windows biases the fit, which is why Hefny et al. regress denoised first-stage estimates in their second stage [23, Sec. 2]. Candidate: state it as a limit of S2 (no fix planned, size unknown).]

- Then the decoder is discarded and (24) continues from the parameters PS leaves; this model is called $\mathcal{M}_{\mathrm{PS}}$:

When S2 is completed the optimization continues minimizing (24)

$$\vartheta_{\mathrm{PS}} = \big(\xi^{(i_1)},\, \theta^{\mathrm{S2}}_{\mathrm{aug}},\, \theta^{(i_1)}_{\mathrm{enc}}\big), \tag{29}$$

- PS changes only the parameters during the first updates: deployed model, criterion and selection unchanged; the added states stay latent

followed by the L-BFGS polish and selection of Section III-D. Here $\xi^{(i_1)}$ and $\theta^{(i_1)}_{\mathrm{enc}}$ are the parameters after S1 and $\theta^{\mathrm{S2}}_{\mathrm{aug}}$ contains the parameters after S2. When S2 is skipped, ϑ_{PS} contains the parameters at the end of S1.
PS has only changed the starting point of minimising V with its initialization phases, not what is being mimimized. The models with and without PS have the same number of parameters, since the decoder is removed.

IV. Interpretability by Orthogonal Construction

- Goal from Section III-A: an estimate of θ_{base} that is unique under stated conditions; names IV-A to IV-C (style profile)
- Contribution share: the third contribution of Section I, in its wording (Introduction)

The split between baseline and network in the augmented model of Section III-D is not unique, and our goal is an estimate of θ_{base} that is unique under stated conditions. To this end, we construct an orthogonal correction of the physical rows (Section IV-A) and state the identification problem of the corrected model (Section IV-B). We then give the condition under which its estimate equals the true combinations, with the scope of this result (Section **??**). This delivers the third contribution of Section I: state-level orthogonality for the identifiable physical-parameter tangent space and its true-parameter recovery condition.

A. Orthogonal correction of the physical rows

- The non-unique split of Section III-A lets the network negate the baseline (Kessels Sec. 5.3.1.3); to first order a term along $\partial f^{\mathrm{n}}_{\mathrm{base}}/\partial\xi$ cancels a change of ξ (this work)
- On the gantry, each direction is an $M(Y)^{-1}$-weighted mass, damping or stiffness force, propagated by RK4 (this work)
- The research plan's penalty trades orthogonality against fit; the construction needs no weight but a baseline linear in its parameters (Györök 2025 Eq. (13), Remark 1; 2026 Sec. 3.2, Eq. (2))
- Regressor: Jacobian of the normalised transition in ξ at ξ_{lin}; range of the ten combinations; no offset column, for two reasons (Györök 2025 Sec. 4, Eqs. (15), (18); D-186; D-192; adapted)
- Fixed tuples of measured positions, differenced velocities, $\bar{x} = 0$, recorded input; data rather than simulated states (Györök 2025 Sec. 3; D-111; D-185 (?))
- Coefficient on the stack, full column rank, orthogonal corrected stack (Györök 2026 Eqs. (8), (11), Lemma 3, Assumption 1; adapted)
- Correction on the physical rows at every step; added-state rows not corrected; coefficient differentiated (Györök 2026 Eqs. (9), (13); D-190; adapted)
- Linearisation point rebuilt per epoch at a detached ξ (Györök 2025 Sec. 4, Remark 2; D-190; adapted)

- Problem: the split between baseline and network is not unique; the network can cancel the baseline (negation)
- Choice: the orthogonal construction of Györök et al.
- Choice: the baseline is not linear in its parameters, so we use its derivative instead: the sensitivity of the one-step map to the ten trained coordinates ξ, taken at the current point ξ_{lin}:

The non-unique split lets the network negate the baseline, Kessels' term for learned terms that partially cancel the first-principles terms [13, Sec. 5.3.1.3]. Let $\xi' \in \mathbb{R}^{10}$ be other values of the training coordinates ξ of (23). The network term $f^{\mathrm{n}}_{\mathrm{aug}} - (\partial f^{\mathrm{n}}_{\mathrm{base}}/\partial\xi)(\xi' - \xi)$ with ξ' then gives, to first order in $\xi' - \xi$, the same one-step map as $f^{\mathrm{n}}_{\mathrm{aug}}$ with ξ. On the gantry, each column of $\partial f^{\mathrm{n}}_{\mathrm{base}}/\partial\xi$ is the RK4 propagation of the acceleration change $-M(Y)^{-1}\big(\frac{\partial M}{\partial\xi_j}\ddot{q} + \frac{\partial C}{\partial\xi_j}\dot{q} + \frac{\partial K}{\partial\xi_j}q\big)$ of (2). A negating term therefore acts as an added mass, damping or stiffness force whose effect on the state depends on Y.

The research plan proposed the orthogonality penalty of Györök et al., whose weight trades orthogonality against fit [25,

Eq. (13), Remark 1]. We instead adopt their later construction, which makes the network term orthogonal to the baseline without a weight [11, Sec. 3.2]. It needs a baseline linear in its parameters [11, Eq. (2)]. Our transition is rational in θ_{base} through $M(Y)^{-1}$, so it has no linear-in-parameters form.

We therefore linearise the normalised transition in its parameters, as Györök et al. do for a nonlinear baseline [25, Sec. 4, Eq. (15)]. We differentiate with respect to ξ rather than the fourteen raw parameters, whose four lost directions (Section II-C) would give the Jacobian four null directions. The map (23) has a nonsingular Jacobian for $m_{\Delta,0} \neq 0$, so the Jacobian in ξ spans the sensitivity directions of the ten combinations θ_{base}. Unlike their extended regressor [25, Eq. (18)], we omit the offset column $f^{\text{n}}_{\text{base}}$ itself, for two reasons. The estimate of ξ moves only within the range of the Jacobian (Section **??**), so the offset would constrain directions that do not move it. A parameter derivative also does not change when the origin of the normalised state moves, whereas the offset does, so the protected directions would depend on the centring μ_x of (15). The sensitivity is

$$\Phi(\xi_{\text{lin}}, \tilde{x}^{\text{n}}, u^{\text{n}}) = \left.\frac{\partial f^{\text{n}}_{\text{base}}\big(\theta_{\text{base}}(\xi), \tilde{x}^{\text{n}}, u^{\text{n}}\big)}{\partial \xi}\right|_{\xi=\xi_{\text{lin}}}, \tag{30}$$

- Choice: only the range of Φ is protected, without the offset column $f^{\text{n}}_{\text{base}}$ of Györök et al.
- Choice: Φ is evaluated on fixed one-step tuples $i \in \mathcal{I}$ from the training data: measured positions, finite difference to get the velocities, added states $\bar{x} = 0$
- The network's physical rows are fitted by the stacked sensitivity in least squares; the remainder is orthogonal to its range:

where $\xi_{\text{lin}} \in \mathbb{R}^{10}$ is the linearisation point and $\Phi : \mathbb{R}^{10} \times \mathbb{R}^6 \times \mathbb{R}^3 \to \mathbb{R}^{6\times 10}$. [**TODO:** Missing: supervisor confirmation of the omitted offset. D-192 (the thesis OBC) gives no reason; the two reasons above are those of D-186 (2026-09-10, reasons 1 and 2), written for the earlier trajectory basis. D-190 records the supervisors' condition as non-overlap with the baseline output, which is the affine range of $[\Phi \;\; f^{\text{n}}_{\text{base}}]$. Candidate: the tangent range with these two reasons and the caveat in Section **??**; parameter uniqueness needs the rank of Φ only [11, Assumption 1]. Confirm with Maarten and Jan.]

We evaluate Φ on fixed reference tuples from the training records, because only the positions of the baseline state are measured. Each tuple holds the positions $q_i = P^{-\top} q_{\text{act},i}$ of a training sample, velocities from a fourth-order central difference of q within the record, $\bar{x}_i = 0$ and the recorded input. The tuples cover every sample at which this difference exists. Each tuple is one step at the recorded input, so the controller does not enter. Györök et al. simulate the baseline when the states are not measured [25, Sec. 3]. In open loop, such a simulation drifts on the free axes (Section III-C). [**TODO:** Missing: reason for data tuples rather than baseline states simulated in closed loop, which do not drift (Section III-C). D-111 rejects only open-loop simulated states, on measurement; D-185 reopened the closed-loop alternative, and no comparison is recorded. Candidate: fixed data tuples do not depend on the trained parameters, so the stack changes only when ξ_{lin} is refreshed (own reading).]

Adapting the coefficient of Györök et al. to these tuples [11, Eqs. (8), (11), Lemma 3], we fit the network's physical rows by the stacked sensitivity in the least-squares sense. As they assume of their regressor [11, Assumption 1], the stack has full column rank:

$$\begin{aligned}
\boldsymbol{\Phi}(\xi_{\text{lin}}) &= \operatorname{col}\big(\Phi(\xi_{\text{lin}}, \tilde{x}^{\text{n}}_i, u^{\text{n}}_i)\big)_{i\in\mathcal{I}},\\
\boldsymbol{f}^{\text{n}}_{\text{aug}}(\theta_{\text{aug}}) &= \operatorname{col}\big(f^{\text{n}}_{\text{aug}}(\theta_{\text{aug}}, \operatorname{col}(\tilde{x}^{\text{n}}_i, 0), u^{\text{n}}_i)\big)_{i\in\mathcal{I}},\\
\theta_{\text{aux}}(\theta_{\text{aug}}) &= \boldsymbol{\Phi}(\xi_{\text{lin}})^{+}\, \boldsymbol{f}^{\text{n}}_{\text{aug}}(\theta_{\text{aug}}),
\end{aligned} \tag{31}$$

so that the corrected stack is orthogonal to the sensitivity:

$$\boldsymbol{\Phi}(\xi_{\text{lin}})^{\top}\big(\boldsymbol{f}^{\text{n}}_{\text{aug}}(\theta_{\text{aug}}) - \boldsymbol{\Phi}(\xi_{\text{lin}})\,\theta_{\text{aux}}(\theta_{\text{aug}})\big) = 0.$$

- Choice: the correction is subtracted from the physical rows at every step; the added-state rows are not corrected:

where $\mathcal{I}$ is the set of tuples, $N = |\mathcal{I}|$, $\tilde{x}^{\text{n}}_i \in \mathbb{R}^6$ and $u^{\text{n}}_i \in \mathbb{R}^3$ are the state and input of tuple i in the coordinates of (15), $\boldsymbol{\Phi}(\xi_{\text{lin}}) \in \mathbb{R}^{6N\times 10}$, $\boldsymbol{f}^{\text{n}}_{\text{aug}}(\theta_{\text{aug}}) \in \mathbb{R}^{6N}$, $^{+}$ is the Moore-Penrose pseudoinverse and $\theta_{\text{aux}} \in \mathbb{R}^{10}$.

As in [11, Eqs. (9), (13)], the coefficient is a function of the network parameters, and every prediction subtracts the sensitivity times the coefficient. We subtract it from the physical rows at every step of the augmented model, at the state and input of that step. The added-state rows are not corrected, because the baseline has no added-state update to be orthogonal to. The correction is

$$f^{\text{n}}_{\text{obc}}(\theta_{\text{aug}}, \tilde{x}^{\text{n}}, u^{\text{n}}) = \Phi(\xi_{\text{lin}}, \tilde{x}^{\text{n}}, u^{\text{n}})\,\theta_{\text{aux}}(\theta_{\text{aug}}), \tag{32}$$

- Choice: the gradient of the training cost V of (24) to the network parameters θ_{aug} includes the path through $\theta_{\text{aux}}(\theta_{\text{aug}}) = \boldsymbol{\Phi}^{+}\boldsymbol{f}^{\text{n}}_{\text{aug}}(\theta_{\text{aug}})$; with θ_{aux} held fixed, the network would be trained on its uncorrected output
- Choice: ξ_{lin} is set to the current ξ at each epoch and held fixed within it

where $f^{\text{n}}_{\text{obc}} : \mathbb{R}^{n_{\text{aug}}} \times \mathbb{R}^6 \times \mathbb{R}^3 \to \mathbb{R}^6$. Differentiating $\theta_{\text{aux}}(\theta_{\text{aug}})$ gives the network the gradient of the corrected term, which a detached coefficient would replace by that of the uncorrected one. [**TODO:** Missing: the planned figure (header): the network's physical stack split into its part in range $\boldsymbol{\Phi}(\xi_{\text{lin}})$ and the orthogonal part, with an inset showing the corrected physical rows, the uncorrected added-state rows and their route back through $f^{\text{n}}_{\text{aug}}$ one step later. Candidate: none drawn yet.]

We set ξ_{lin} to the current ξ at each epoch boundary, rebuild $\boldsymbol{\Phi}(\xi_{\text{lin}})$ there and hold it for the epoch. No gradient therefore reaches ξ through Φ. This adapts the per-epoch refresh that Györök et al. allow in their linear case [25, Remark 2] to the linearisation point. For a nonlinear baseline, they update this point at every evaluation of the objective or fix it at the nominal parameters [25, Sec. 4]. The stacked sensitivity changes little within an epoch, so a rebuild at every evaluation would mainly add cost. [**TODO:** Missing: (a) a reported measurement for

the slow change of the stacked sensitivity (D-190 measured its principal angles at 1 and 20 percent displacement; no section reports them). Candidate: one sentence with the numbers in a new appendix subsection. (b) The reason against a linearisation point fixed at the start. Candidate: the start need not be the nominal point that Györök et al. argue for, since training may start away from the true combinations (own reading).]

B. Identification of the corrected model

- Need: the identification problem of Section III-D must include the correction (32); trained parameters, coordinates and cost stay (2026 Eq. (22); code)
- Problem display: cost and changed physical-row line, the other constraints by number; ξ_{lin} held fixed and refreshed per epoch, θ_{aux} re-solved at every evaluation (D-190 items 5, 6; code)
- Per phase: S1 fits the corrected residual; S2 does not see the correction but changes θ_{aug}; Adam updates; validation uses the last epoch's basis; selection rebuilds the basis; the polish keeps it (D-198; code)
- $\mathcal{M}_{\mathrm{OBC}}$ named, input and output as in Section III; compared with $\mathcal{M}_{\mathrm{PS}}$ in Results (THESIS-RESULTS steps 5, 6; notation (?))
- The identification problem of Section III-D gets the correction in the physical-row update; trained parameters and cost are unchanged:

The identification problem of Section III-D must now include the correction in the physical-row update. The trained parameters $\vartheta = (\xi, \theta_{\mathrm{aug}}, \theta_{\mathrm{enc}})$ and the cost V are unchanged:

$$\begin{aligned} &\min_{\vartheta} \; V(\vartheta) \quad \text{s.t.} \\ &\tilde{x}^{\mathrm{n}}_{k+1|\tau} = f^{\mathrm{n}}_{\mathrm{base}}\big(\theta_{\mathrm{base}}(\xi), \tilde{x}^{\mathrm{n}}_{k|\tau}, \hat{u}^{\mathrm{n}}_{k|\tau}\big) + f^{\mathrm{n}}_{\mathrm{aug}}\big(\theta_{\mathrm{aug}}, \hat{x}^{\mathrm{n}}_{k|\tau}, \hat{u}^{\mathrm{n}}_{k|\tau}\big) \\ &\qquad\qquad - f^{\mathrm{n}}_{\mathrm{obc}}\big(\theta_{\mathrm{aug}}, \tilde{x}^{\mathrm{n}}_{k|\tau}, \hat{u}^{\mathrm{n}}_{k|\tau}\big), \\ &\text{the other constraints of (24),} \end{aligned} \tag{33}$$

- θ_{aux} is solved again at every evaluation of V; ξ_{lin} is refreshed per epoch
- The PS stages, optimiser and selection rule are reused; $\mathcal{M}_{\mathrm{OBC}}$ is $\mathcal{M}_{\mathrm{PS}}$ with the correction

where ξ_{lin} in (32) is held fixed during each update and refreshed per epoch (Section IV-A), and θ_{aux} is solved by (31) from the current θ_{aug} at every evaluation of V.

The optimiser, the stages of Section III-E and the selection rule of Section III-D are reused. Each S1 update fits the residual of the corrected model. It changes only θ_{p} and $\theta^{a}_{\mathrm{enc}}$. S2 fits $g^{\mathrm{n}}_{\mathrm{aug}}$, which the correction does not touch. It changes θ_{aug}, so θ_{aux} changes at the next evaluation of V. The Adam updates of V, during S1 and in S3, change every entry of ϑ. Validation uses the basis of the last epoch. After the best Adam checkpoint is restored, the basis is rebuilt at its ξ. The polish then changes every entry of ϑ with this basis, under the acceptance rule of Section III-D.

We call $\mathcal{M}_{\mathrm{OBC}}$ the model $\mathcal{M}_{\mathrm{PS}}$ with the correction (32), identified by (33). Its input is the actuator force u_{act} and its output the actuator positions q_{act}. It starts as $\mathcal{M}_{\mathrm{PS}}$ does, with the first basis built at $\xi = 0$. Section VI compares $\mathcal{M}_{\mathrm{OBC}}$ with $\mathcal{M}_{\mathrm{PS}}$ on the ten combinations and on accuracy. [**TODO:** Missing: model notation. Candidate: the calligraphic $\mathcal{M}_{\mathrm{OBC}}$, since plain M clashes with $M(Y)$ (README open conflict on model names).]

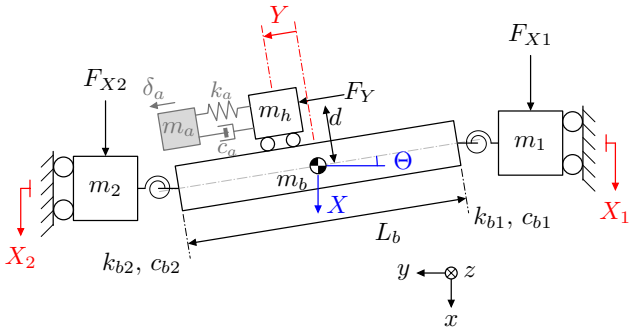


Fig. 6: Simulated gantry with a payload-mounted absorber. The light-gray elements $(m_a, k_a, c_a, \delta_a)$ are absent from the baseline.

V. Experiment Design

A. System and data

- System: LPV baseline plus a payload absorber (50 % of the head mass, 150 Hz, $\zeta = 0.03$) plus tanh Coulomb rail friction, in closed loop with controller K1 (Fig. 6)
- Choice: absorber and friction only in the system, as known missing dynamics
- Choice: output noise at the measured machine level; noise floor = noisy record minus its noise-free twin
- Excitation: multisine of 106 to 297 Hz
- Types of motion: standstill at a fixed Y, slow Y sweeps, X-Y Lissajous paths, S-curve point-to-point moves, sine-shaped
- Training (18 records): fits the model; Y within ± 0.30 m, accelerations up to 75 % of the machine maximum
- Validation (6 records): only selects the checkpoint and the black-box size; never reported
- Test (12 records): every reported number
- Interpolation (I1 to I6): inside the training range, with new multisine phases, new Y positions and new move sequences (sine and S-curve routes, X-only and Y-only moves, long strokes)
- Extrapolation (E1 to E6): each differs from its I record in one property outside the training range: Y position (E1, E2, up to 0.39 m), acceleration (E3, 100 %), velocity (E4, 1.95 m/s), setpoint using movement from machine (E6), a changed controller (E5, not done currently)

B. Compared models

- Baseline: LPV at the true and at the detuned parameters; LTI frozen at $Y = 0$
- Augmented: static ($n_a = 0$), plain training, PS initialisation ($n_a = 2$ and 8), unconstrained against OBC

- Choice: start 10 % detuned for the combinations, the same direction in every run
- Choice: black box = SUBNET on the closed loop $(r, f) \to e$, size selected from 11 configurations on validation

C. Training and evaluation

- Closed-loop training, window 0.1 s, encoder history 29 samples, network 2×16 tanh
- Choice: one seed per run (compute limit)
- Metric: simulation error RMS per axis in m, over the whole test record
- Validation only selects checkpoints; every reported number is on the test set

[**TODO:** Define the two overlaps of Table II (moved from the removed Section IV-C). On one-step tuples of I1 to I6 at $\bar{x} = 0$, with Π the orthogonal projector onto range $\boldsymbol{\Phi}$ of (31): learned-term overlap $\|\Pi \boldsymbol{f}^{\mathrm{n}}_{\mathrm{aug}}\|_2 / \|\boldsymbol{f}^{\mathrm{n}}_{\mathrm{aug}}\|_2$, with $\boldsymbol{\Phi}$ at the trained coordinates and, for OBC, $\boldsymbol{f}^{\mathrm{n}}_{\mathrm{aug}}$ after the correction (32); missing-dynamics overlap $\|\Pi\varepsilon\|_2 / \|\varepsilon\|_2$, $\varepsilon = \tilde{\boldsymbol{x}}^{\mathrm{n}}_{+} - \boldsymbol{f}^{\mathrm{n}}_{\mathrm{base}}(\xi^{\star})$, with $\boldsymbol{\Phi}$ at the true coordinates $\xi^{\star}$ and $\tilde{\boldsymbol{x}}^{\mathrm{n}}_{+}$ the next states of the tuples.]

- Generate closed-loop records from the physics baseline plus a truth-only flexible absorber, retaining exact states solely for diagnostics
- Use multisine excitation that covers the absorber pole and anti-resonance while respecting position, velocity, yaw, and force limits
- Assign independent random multisine phases to training, validation, and test records so equal spectra do not duplicate trajectories
- Include multisine-free cycloidal motion profiles matching the measured machine motion to test the operational distribution shift
- Use only the corrected all-channel Coulomb dataset matched to the frictionless excitation; the earlier mismatched-amplitude merge is superseded
- The set-valued friction law requires a velocity threshold below which a rail is treated as stuck. This is set to the Stribeck velocity, $v_s = 2.25 \cdot 10^{-3}$ m/s, taken from Lee et al. [26], Table 5, identified on a ball-screw feed drive with a THK SSR20XW guide (Sec. 5.1). Stribeck velocities reported in the literature span 10^{-5} to 10^{-2} m/s [26, Sec. 5.2], so the chosen value sits mid-range [**TODO:** Conflicts with D-204, where the stick band is `V_EPS = (cc1+cc2)/m_total*ts` labelled THEORY: leine1998, about $3.3 \cdot 10^{-5}$ m/s before the 3 to 10x margin. Confirm which threshold the reported runs use, and whether the second citation (Stribeck range) is the same paper.]
- State that output noise is currently disabled; add it only if a level and measurement location can be estimated from Jasper, ILC data, or stationary records
- State the sample rates, record lengths, number of windows, and total independent motion time rather than quoting overlapping-window count as data volume
- Decision (RK4 value, moved here from Section II): one reproducibility sentence giving the RK4 step, equal to the sampling period after downsampling the 20 kHz records to 4 kHz, with one substep (entry script: `fs_new=4000`, `up_sample=1`). Confirm against the `config.json` of each reported run, since earlier checkpoints may have used another `up_sample`
- Figure (reuse, no separate asset): let the rebuilt training-objective time strip also identify independent phases, short training windows, encoder history, and the long evaluation horizon

[**TODO:** Is the hidden linear absorber sufficient simulation complexity, or is a truth-only cubic spring required to demonstrate nonlinear augmentation?] [**TODO:** discuss noise model and how we filter noise in the setup]

D. Real Telica data and split

- Confirm with Jasper and Dragan which reconstructed ETEL setpoint-generator or ILC-reference motions are operationally representative, including minimum/maximum move, long stroke, acceleration, and dwell
- Use the supervisor-defined split of 11 training, two validation, and two test operating points at 20 kHz
- Convert rail positions and currents to logical metres and Newtons; form velocity states by central differences because direct velocity is not logged
- Keep all iterations at one operating point in the same split to prevent identical reference trajectories leaking across sets
- Describe the X and Y operating-point grid and show whether training covers the interpolation region occupied by both test points
- Use iteration 0 for controller checks and later ILC iterations for distinct feedforward conditions without treating them as independent trajectories
- Mark real-data training and held-out evaluation as planned until a final reproducible run and force-calibration convention exist
- Figure (new): 2D operating-point grid with train, validation, and test locations, operating-range boundary, and trajectory classes; add a 3D or pairwise state-input coverage view only if it reveals coverage hidden by the 2D map

[**TODO:** Freeze the approved operational trajectory set with Jasper and Dragan; should direct drive velocity be requested, and does a third coverage coordinate add information beyond the 2D grid?]

E. Training protocol, comparators, and metrics

- Match records, neural seeds, windows, optimiser budget, and checkpoint criterion across baseline-only, unconstrained, tangent-OBC, affine-OBC, and black-box arms; treat multisine phase seeds separately
- Use physical RMS as the primary candidate metric for tracking and a fixed 200–400 ms post-move settling

interval; use NRMS, BFR, or spectra only for a distinct supporting question
- Report physical combination error, projection overlap, added-state ablations, and encoder-init sensitivity alongside predictive metrics
- Values referenced from Section III (moved 2026-10-06), each with an honest basis (derived, inherited, tuned, compute-limited): encoder history n, window length n_f and stride, number of additional states $n_{\bar{x}}$, network width and depth, model rate
- Compared model (moved from Section III-A, 2026-10-11): for $n_{\bar{x}} = 0$ the neural network only corrects the six baseline states, giving the static parallel augmentation of [8, Table 1]; used in the $n_{\bar{x}}$ comparison (`THESIS-RESULTS.md` step 4, run R6)
- Free-run validation (referenced from Section III-C): whole records, encoder at the record start and $x^c = 0$ once, so model error accumulates over the record unlike in a training window; state the record-length-weighted RMS used for checkpoint selection
- Decision (moved from Section II-C): joint-estimation arms start from the nominal φ of the reference implementation detuned by 10% per combination (`combo_init_detune`, HEURISTIC, D-191), with the nominal-parameter prior disabled (D-193); state it as a run setting with its reason
- A parameter prior bounds the drift of the combinations but does not remove the negation flat direction of Section IV; it is off in every arm (D-193)
- Practical identifiability (referenced from Section II-C): report the rank of the stacked one-step transition Jacobian with respect to φ on the training data (rank 10, nullity 4 in θ per the reduced-block docstring), from a saved artefact rather than the docstring
- Scope (from Section II-C): the structural result concerns the ideal, noise-free baseline model; whether the closed-loop records excite all ten combinations is checked here; the augmented model, where the network can absorb part of the baseline, is treated in Section IV; the encoder does not affect the result, as it only estimates the initial state of a simulation, not the model
- Value (moved from Section II-C): the margin $\rho = 0.99$ in (23), a HEURISTIC margin inside the admissible region (D-229)
- Metric (moved from Section II-C): relative combination error in θ_{base}, with the normalisation of the small signed m_Δ stated explicitly
- State per arm whether raw or reduced coordinates are trained; exact-model parameter recovery is an internal implementation check, not a result (moved from Section II-C)
- Give the black-box model validation-selected capacity and horizon so it is a capability comparison rather than an artificially weakened baseline
- Evaluate held-out position/phase interpolation, ASMPT-relevant multisine-free motion, controller transfer, and operating-range extrapolation as separate rungs
- Predeclare expectation, acceptance criterion, verdict, and mechanism for every result rather than presenting isolated scores

[**TODO:** Decide the primary metric, credible controller-transfer arm, and feasible neural-seed count; do not pool initialization, noise, phase, and operating-point uncertainty.]

F. Reproducibility and implementation checks

- Verify controller reconstruction, coordinate transforms, normalisation, RK4 transition, and LFR equivalence before training
- Use the same controller, data conversion, and rollout entry point for training, validation, and checkpoint selection
- Controller rate (moved from Section III-C, 2026-10-06): the records are generated with the controller at 20 kHz, the training loop re-discretises it at the 4 kHz model rate (D-141); the residual form of Section III-C equals the shared-reference loop only for a compatible controller, signals and initial state
- Compatibility check for that assumption: run the re-discretised controller on the recorded $r - y$ and compare its feedback force with the recorded one; a rollout with $\hat{y} = y$ gives $\hat{u} = u$ by construction and checks nothing
- Controller mismatch: the records use the controller at 20 kHz, training re-discretises it at 4 kHz, so the residual form of Section III-C is approximate and part of the measurement noise is not removed [**TODO:** Missing: size of the controller mismatch. The records use the controller at 20 kHz, training re-discretises it at 4 kHz. Candidate: the compatibility check of the Setup bullet, reported as the force difference against the recorded u_{act}.]
- Record configuration, software precision, seed, dataset identifier, and checkpoint provenance for every reported arm
- Keep oracle states and truth parameters out of the trainable pipeline and label every diagnostic that uses them
- Figure (existing, appendix candidate): use `controller-chain-v1` only as an audit trail for controller reconstruction and discretisation, not as a main method figure

[**TODO:** Which implementation checks belong in the main paper, and which should move to the appendix or repository documentation?]

VI. Results

A. Baseline form

- Criterion: LPV error flat over Y, LTI error growing away from $Y = 0$ (Fig. 7)
- On X the LTI error is 1.5 to 3.7 times the LPV error, growing with $|Y|$; on Y both are equal
- Y scheduling is needed for the X axes
- Expectation and criterion: the LPV-LFR and original physics equations agree numerically across the payload range before model-error comparisons are meaningful

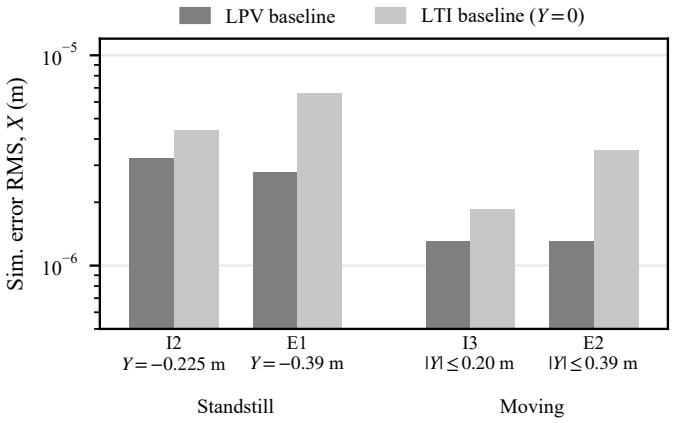


Fig. 7: Simulation error RMS on the X axes of the LPV baseline and the LTI baseline at $Y = 0$, both at the true parameters.

- Verdict and reason: report inertia invertibility, LFR/RK4 equivalence, and the remaining baseline error caused by the truth-only absorber
- Figure: baseline and oracle output errors versus time for representative payload positions; supports the claim that the omitted mode is systematic and position dependent
- Table: baseline free-run physical RMS for LPV and frozen-LTI models across held-out Y; tests the research plan's Aspect 1 rather than assuming LPV superiority
- Expectation and criterion: direct and residual closed-loop forms match at numerical precision when controller assumptions hold
- Evidence for Section III-C (moved 2026-10-06): on the thesis data, open-loop rollouts accumulate position error on the free axes X and Y, closed-loop rollouts do not; re-measure, do not reuse the older D-142 numbers
- Figure: recorded input, controller correction, and model error over one rollout; supports the interpretation that feedback can suppress visible output error while the open-loop model remains wrong

[**TODO:** The LPV-versus-frozen-LTI comparison is required by the plan but may not yet have a final matched result; run it or narrow Aspect 1 explicitly.]

B. Augmentation structure

- Criterion: error below the baseline and below the model without added states (Table I)
- PS initialisation lowers the Y error about 9 times; plain training and $n_a = 0$ stay near the baseline
- The error in the multisine band disappears with PS initialisation, not with plain training (Fig. 8)
- The learned pole is at 212 Hz, the absorber pole of the system at 212 Hz; $n_a = 8$ adds nothing
- Without added states ($n_a = 0$) the Y error stays at the baseline level
- Time trace of the augmented model and the system (Fig. 9)

TABLE I: Simulation error RMS per axis, mean over test records I1 to I6.

Model	X_1 (m)	X_2 (m)	Y (m)
Baseline, true θ	$1.64 \cdot 10^{-6}$	$1.97 \cdot 10^{-6}$	$1.31 \cdot 10^{-5}$
Baseline, detuned θ	$2.99 \cdot 10^{-6}$	$2.28 \cdot 10^{-6}$	$1.56 \cdot 10^{-5}$
Static parallel, n_a=0	$1.33 \cdot 10^{-6}$	$1.42 \cdot 10^{-6}$	$1.24 \cdot 10^{-5}$
Plain training, n_a=2	$1.18 \cdot 10^{-6}$	$1.26 \cdot 10^{-6}$	$1.06 \cdot 10^{-5}$
PS init., n_a=2	$1.09 \cdot 10^{-6}$	$1.21 \cdot 10^{-6}$	$1.39 \cdot 10^{-6}$
PS init., baseline part only	$1.94 \cdot 10^{-6}$	$1.77 \cdot 10^{-6}$	$1.35 \cdot 10^{-5}$
PS init., n_a=8	$1.04 \cdot 10^{-6}$	$1.19 \cdot 10^{-6}$	$1.44 \cdot 10^{-6}$
PS init., detuned θ	$1.08 \cdot 10^{-6}$	$1.20 \cdot 10^{-6}$	$1.44 \cdot 10^{-6}$
Augmented (OBC), detuned θ	$1.14 \cdot 10^{-6}$	$1.29 \cdot 10^{-6}$	$1.49 \cdot 10^{-6}$
Black-box model	$1.85 \cdot 10^{-5}$	$2.11 \cdot 10^{-5}$	$2.33 \cdot 10^{-5}$
Noise floor	$9.00 \cdot 10^{-9}$	$9.63 \cdot 10^{-9}$	$3.19 \cdot 10^{-9}$

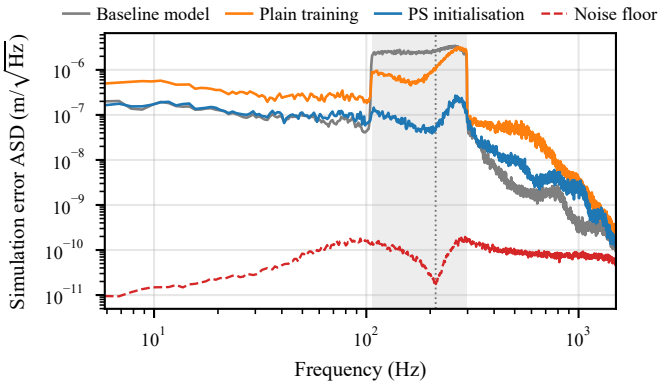


Fig. 8: Amplitude spectral density of the Y simulation error on test records I1 and I2. Shaded: multisine band. Dotted: absorber pole.

- Expectation and criterion: augmentation must reduce held-out long-horizon error, not only the short training loss, relative to the same encoder-initialised baseline
- Verdict and reason: report the best validated improvement and the gap between window fit and free-run accuracy caused by integrated velocity bias
- Figure: validation RMS against optimiser updates for baseline and augmented arms, with window and 12 s metrics; supports or rejects objective alignment
- Figure: error spectra before and after augmentation with absorber pole and anti-resonance marked; tests whether the learned correction targets the missing band
- Figure: clamp each additional state and the full added-state vector to zero during rollout; quantifies what the trained latent dynamics add or harm
- Table: standard model, no-added-state ablation, zero-initialised latent path, and longer encoder history; connects structure changes to predictive benefit
- State the observed weak or harmful added-state contribution as a result if the final run does not reverse it, rather than inferring usefulness from state magnitude

[**TODO:** Are state Gramians useful for explaining each added state's contribution, or would intervention-based ablation

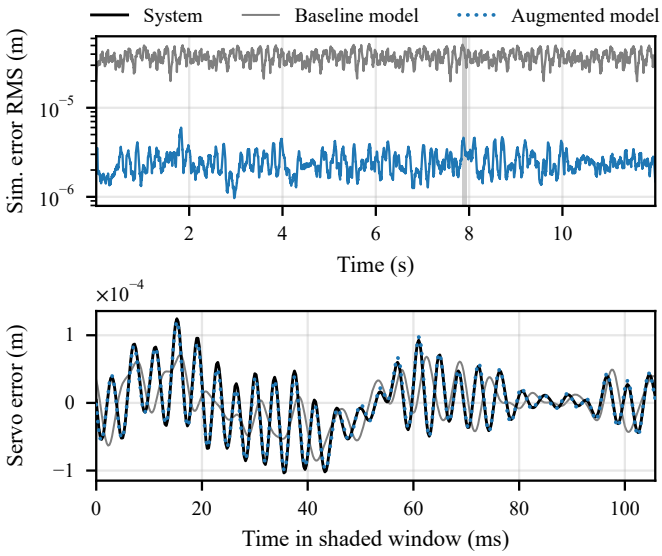


Fig. 9: Test record I1, Y axis. Top: moving RMS of the simulation error. Bottom: servo error in the shaded window.

be more direct and less linearisation-dependent?]

C. Interpretability

- Criterion: OBC drifts less than the unconstrained arm at equal test RMS (Table II)
- Not met: the parameters drift in both arms, OBC not less
- OBC keeps the learned term out of the parameter directions: overlap 0.12 to 0.13 against 0.91 to 0.99
- Cause: 59 % of the missing dynamics lies in the parameter directions

- Evaluation on the validation-selected best checkpoints of R1/R2 (OBC, detuned/correct start) and R4/R5 (unconstrained, detuned/correct start): the following diagnostics require no additional training runs; all reported predictive scores use the test set.
- Report normalised inner products with the baseline sensitivity space and the baseline-output overlap ρ_f (Eq. (**??**)), both on the $x_a = 0$ reference slice and with active added states along held-out test rollouts. Use the same normalised coordinates and stacking convention, and distinguish reference-slice orthogonality from observed overlap away from that slice.
- Report the ten physical combination errors relative to truth, their aggregate relative-error norm, and displacement from each start; compare OBC with unconstrained estimation for both starts.
- Report full augmented-model and standalone-baseline test prediction RMS using each checkpoint's fitted physical parameters, against the per-record noise floor, to assess physical-model usefulness alongside predictive accuracy.
- Interpret overlap and agreement between the two starts as empirical diagnostics of the decomposition. They do not prove uniqueness or true-parameter recovery; any uniqueness statement must retain the assumptions and scope of the derivation in Section **??**.
- Expectation and criterion: OBC should reduce physical combination error without materially worsening held-out predictive fit relative to unconstrained joint estimation
- Verdict and reason: the completed arms reach 7.20% no projection, 7.69% tangent OBC, and 8.47% unfinished affine OBC from a common 9.49% start, so aggregate recovery does not establish OBC superiority
- Figure: validation fit versus combination error over training for all three arms; supports the claim that near-equal fit permits materially different physical parameters
- Figure: per-parameter error at matched budget; shows which combinations projection protects and which near-flat directions redistribute error
- Table: basis rank, condition, overlap, final and minimum parameter error, and validation RMS; links algorithm health to the recovery verdict
- Figure: ρ^* and predicted versus observed parameter bias by singular direction; supports the conditional result that true recovery is limited by non-orthogonal missing physics
- Explain why the 4.96% predicted floor accounts for part of the residual squared error but does not match the observed per-parameter pattern or prove convergence

[**TODO:** Should the incomplete affine arm be retained, rerun to a matched budget, or reported only as exploratory evidence?]

TABLE II: Relative error of the ten combinations, unconstrained and OBC, from the detuned and the correct start.

	Detuned start			Correct start	
Rel. error (%)	Start	Unconstr.	OBC	Unconstr.	OBC
$k_{b,\Sigma}$	−10.0	−2.3	−6.2	−9.2	−2.7
c_{g1}	+10.0	+35.1	+45.0	+16.4	+28.3
c_{g2}	+10.0	+44.7	+44.1	+29.4	+35.3
c_y	+10.0	+41.5	+12.6	+30.0	−6.0
$c_{b,\Sigma}$	−10.0	+15.2	+19.9	+28.8	+34.7
m_h	−10.0	−3.3	−3.6	−2.2	−2.2
m_Σ	+10.0	+2.4	+2.5	+3.1	+2.0
m_Δ	+10.0	+33.0	+44.5	+14.4	+34.5
J_{eff}	−10.0	−0.8	−1.9	+3.8	+2.6
d	−10.0	−3.2	+10.4	−9.0	+14.9
Norm	31.6	79.4	81.7	57.1	68.7
Test RMS ($\times 10^{-6}$ m)		1.24	1.31	1.23	1.30
Learned-term overlap		0.99	0.12	0.91	0.13
Missing-dynamics overlap			0.59		
Learned pole (Hz)		212	213	212	214
Learned damping ζ		0.049	0.052	0.051	0.052

D. Generalisation

- Criterion: at least as good as the black-box model on I, better on E (Fig. 10)
- Met: the augmented model is below the detuned baseline on all records, and 10 to 30 times below the black-box model on I and E
- During motion about 10^{-6} m remains, 100 times above the noise floor

- Expectation and criterion: improvement must progress from held-out position/phase interpolation to ASMPT-relevant motion, controller transfer, and quantified operating-range extrapolation without retuning
- Verdict and reason: separate completed simulation evidence from pending Telica results and explain any failure by excitation coverage, initialisation, or objective mismatch
- Figure: tracking overview plus a dedicated 200–400 ms settling zoom for baseline, augmented model, and black box on an approved operational trajectory
- Table: primary RMS, fixed-window settling RMS, parameter availability, and training data for baseline, augmented, and black-box models on every completed validation rung
- Figure or table: train with one controller and evaluate the frozen models with a credible changed controller; pair this with a plant-domain check for feedback masking
- Figure: error versus multisine amplitude, stroke, acceleration, scheduling position, or state-input distance from the training set; quantify extrapolation rather than calling all held-out data extrapolation
- Report the black box's early improvement and later free-run degradation as a negative comparator result if no stable matched run supersedes it
- Aggregate the final learned-model comparison across neural initialization seeds and show a representative, not best, time trace; report phase realizations separately

[**TODO:** Freeze which validation rungs are complete enough for a final claim; omit rather than speculate about unfinished controller, noise, or hardware experiments.]

VII. Discussion

A. Answers to the four research aspects

- Aspect 1: state whether explicit Y scheduling earns measurable predictive value over frozen LTI baselines
- Aspect 2: distinguish the expressive value of dynamic augmentation from the observed trainability of its added states
- Aspect 3: conclude that OBC enforces a unique local split but true parameter recovery is conditional on the real discrepancy's orthogonality and optimiser convergence
- Aspect 4: compare simulation interpolation, motion-profile transfer, amplitude extrapolation, real-data evidence, and the black-box result without merging them into one score
- Treat controller transfer as the industrial digital-twin use case and as a closed-loop distribution shift distinct from operating-range extrapolation
- Interpretability matters when physical parameters support controller redesign or health monitoring, but its practical value must be tied to a named downstream decision
- Figure plan: introduce no new figure; interpret the architecture, OBC, and result figures already presented instead of adding a summary graphic

[**TODO:** Which result is the primary answer if predictive accuracy and parameter recovery favour different arms?]

B. Limitations and lessons

- Closed-loop feedback improves numerical trainability but can hide plant mismatch, suppress orthogonality overlap, and reward a model that is poor in open loop
- Short windows underweight velocity DC bias on free integrators; varying encoder and rollout length is therefore a structural sensitivity test, not routine tuning
- One strengthened absorber, currently noiseless simulation, a finite neural-seed campaign, and truth-only dry friction bound claims about nonlinear, noisy, and dissipative missing dynamics
- State the final seed count and corrected-friction evidence here; do not retain development-era caveats after the final campaign changes them
- Central-difference velocities, force calibration, and the train-validation-test coverage grid bound the credibility of any real-data conclusion
- The research plan expected validated real-machine generalisation; any missing Telica result is a departure to disclose, not replace with design documentation
- Augmentation versus residual-FRF fitting remains a trade between trajectory-domain nonlinear prediction and a simpler frequency-domain account of missing dynamics
- Pointwise orthogonality is impossible for any nonzero missing force on this gantry: the parameter sensitivity of the three velocity rows has rank 3 at every reference sample (worst $\sigma_3/\sigma_1 = 5.7 \cdot 10^{-4}$, earlier absorber dataset), so $J^\top \Delta^* = 0$ can only arise from cancellation over the record (Section IV-C)
- Coulomb friction in the truth violates the recovery condition by construction: the c_{g1} sensitivity is proportional to $\dot{X}_1$ and the friction is $-c_{c1}\,\mathrm{sign}(\dot{X}_1)$, so their inner product is $-c_{c1}\sum|\dot{X}_1| \neq 0$ for any moving record (D-204); on T_AF, OBC gives a unique split, not unbiased combinations

[**TODO:** Should a cubic-spring benchmark, noise study, and Monte Carlo analysis be completed, or should the thesis explicitly narrow its claims to the measured linear absorber case?]

C. Implications and future work

- Design excitation so $J^\top \Delta^* = 0$ when true parameter recovery, rather than only a unique decomposition, is required
- Separate learning rates for physical and neural parameters and compare convergence against the predicted nonzero OBC bias floor
- Replace the short-horizon objective or couple windows so integrated velocity bias is visible during training
- Repair and rerun the Coulomb benchmark before claiming that the method learns a legitimate nonzero-mean residual
- Extend evaluation beyond the multisine range and use machine-relevant trajectories to delimit safe interpolation and extrapolation

[**TODO:** Which future-work item is a prerequisite for the thesis claim and which genuinely belongs after graduation?]

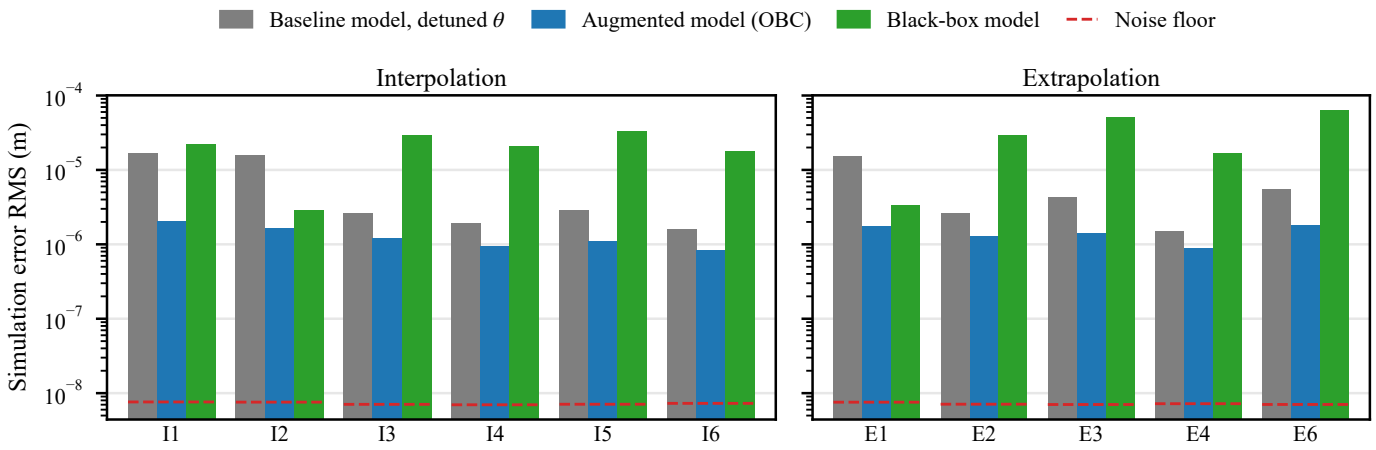


Fig. 10: Simulation error RMS per test record, mean over X_1, X_2, Y.

VIII. Conclusions and Discussion

A. Conclusions

- Aspect 1: LPV scheduling lowers the error on X, increasingly away from $Y = 0$
- Aspect 2: added states with PS initialisation learn the missing mode; plain training does not
- Aspect 3: OBC keeps the learned term out of the parameter directions; the parameters still drift
- Aspect 4: better than the black-box model on interpolation and extrapolation
- Black box currently doesnt perform well on test sets.
- Recommendation?

- Answer whether LPV-LFR dynamic augmentation improves dual-gantry settling prediction over the physics baseline and black-box comparator
- State whether the learned additional states contributed measurable missing dynamics under the final validated configuration
- Conclude that state-level OBC prevents local tangent overlap by construction, while true parameters are recovered only when the physical discrepancy is orthogonal on the chosen excitation
- Give the final quantitative trade between prediction error and physical combination error on held-out data
- State whether the evidence covers relevant ASMPT motion, settling, controller transfer, or only a narrower simulation validation ladder
- Close with the practical recommendation for when ASMPT should use the augmented model and when the baseline or a simpler residual model is safer
- Figure plan: no conclusion figure; reserve the space for the final quantitative answer and limitations

[**TODO:** Rewrite every conclusion bullet after freezing the final result table; no planned or diagnostic result may appear as achieved.]

B. Discussion

- Why the parameters drift despite OBC (missing-dynamics overlap)
- What the black-box comparison does and does not show
- The error during motion is not captured, likely friction??
- Single seed, simulation only, no controller transfer
- OBC holds only at $\bar{x} = 0$; current definition doesn't take into account the additional states entering the output the next update step.
- Need to fix: current encoder is initialized with true parameters
- Future work: friction, excitation for parameter recovery, real data

- Why all states routing.
- why not use the encoder for the orthogonality each epoch instead of finite difference
- why not use the positions since we measure them –¿ train nn and encoder in combination? not full answer
- black box not fair comparison, no init (BLA) no controller

Thesis

Make a new folder for figures and code of the figures for the paper, and the final dataset that will be used.

* laten zien met parameters samenvoegen huidige methode niet correct ik moet iets van jacobian of iets laten zien * ik heb nog steeds niet netjes uitgewerkt voor LPV-LFR met hoe ik het heb omgeschreven die n m ik kan niet zo goed zeggen hoe het heet * verbeterde model met additional states laten zien als ik ze weg haal wat ze toevoegen = op nul zetten. ik weet niet of andere phase multisine validation en test genoeg laat zien hoe goed het werkt. * Duidelijk verantwoorden controller. vooral met integrator problem van de plant en dus X en Y geen stijfheid K = 0 daar. kleine integratie error propegeert gwn door, dit zei maarten ook denk dat dit de error is. * velocity central difference, why not get direct? * Zijn Gramians nuttig om per state te laten zien? * Multisine phase uitleggen kunnen we ook bronnen voor gebruiken. dat verschillende train validation en test verschillende multisine phase hebben. * Met alleen simulatie is huidige simulatie niet genoeg, cubic spring toevoegen?? * benefit van mijn closed-loop training method

is ook dat ik geen encoder / states er voor hoef te initializen. kessels wel referencen. figuur gebruiken. * Jacobian additional states * 2d / 3d grid wat jan zei van de data space of ik alles vang van het gebied van telica. * Montecarlo simulatie zoals jan? * For parameter recovery I had trajectories similar to the paper, currently not used for joint estimation? * How well do we extrapolate outside multisine range? what does jan do?

Verdedigen

* waarom geen basis functions * waarom niet zoals kessels: vooral andere literatuur kunnen zeggen waarom niet dat gebruikt, moet dat? * wat is het voordeel van interpreteerbaarheid?–¿ controller maken, monitor health? * waarom niet frf of mass lijn fitten voor de missende dynamica * controller surpressses / hides issues such as orthogonality overlap. Byproduct of closed-loop. * Hoe verdedig ik augmentation over residual frf fit?? * Hoeveelheid data kunnen verdedigen * Huidige nf window and encoder window, should i run for different lengths? * Orthogonality definition and that we orthogonalize additional states with respect to the baseline := 0. additional states propogate, not included in our definition. * Kijken decisions.md * do i need to justify controller can i say what i got from my supervisor, how to approach these kind of questions?

REFERENCES

[1] International Roadmap for Devices and Systems (IRDS), "International roadmap for devices and systems: 2022 edition executive summary," IEEE, Tech. Rep., 2022.

[2] T. Workman, L. Mirkarimi, J. Theil, G. Fountain, K. M. Bang, B. Lee, C. Uzoh, D. Suwito, G. Gao, and P. Mrozek, "Die to wafer hybrid bonding and fine pitch considerations," in *2021 IEEE 71st Electronic Components and Technology Conference (ECTC)*. IEEE, 2021, pp. 2071–2077.

[3] I. García-Herreros, X. Kestelyn, J. Gomand, R. Coleman, and P.-J. Barre, "Model-based decoupling control method for dual-drive gantry stages: A case study with experimental validations," *Control Engineering Practice*, vol. 21, no. 3, pp. 298–307, 2013.

[4] C. S. Teo, K. K. Tan, S. Y. Lim, S. Huang, and E. B. Tay, "Dynamic modeling and adaptive control of a h-type gantry stage," *Mechatronics*, vol. 17, no. 7, pp. 361–367, 2007.

[5] P. Lambrechts, M. Boerlage, and M. Steinbuch, "Trajectory planning and feedforward design for electromechanical motion systems," *Control Engineering Practice*, vol. 13, no. 2, pp. 145–157, 2005.

[6] J. Hwang and S. D. Noh, "Digital twin-based optimization of operational parameters for cluster tools in semiconductor manufacturing," *IEEE Access*, vol. 12, pp. 122 078–122 100, 2024.

[7] J. Willard, X. Jia, S. Xu, M. Steinbach, and V. Kumar, "Integrating scientific knowledge with machine learning for engineering and environmental systems," *ACM Computing Surveys*, vol. 55, no. 4, pp. 1–37, Nov. 2022.

[8] J. H. Hoekstra, B. M. Györök, R. Tóth, and M. Schoukens, "Learning-based augmentation of first-principle models: A linear fractional representation-based approach," 2026, submitted to Automatica.

[9] G. E. Karniadakis, I. G. Kevrekidis, L. Lu, P. Perdikaris, S. Wang, and L. Yang, "Physics-informed machine learning," *Nature Reviews Physics*, vol. 3, no. 6, pp. 422–440, Jun. 2021.

[10] M. Raissi, P. Perdikaris, and G. E. Karniadakis, "Physics informed deep learning (part i): Data-driven solutions of nonlinear partial differential equations," *arXiv preprint arXiv:1711.10561*, 2017.

[11] B. M. Györök, M. Schoukens, T. Péni, and R. Tóth, "Orthogonal-by-construction augmentation of physics-based input-output models," *IFAC Journal of Systems and Control*, vol. 35, p. 100376, 2026.

[12] J. P. Noël and J. Schoukens, "Grey-box state-space identification of nonlinear mechanical vibrations," *International Journal of Control*, vol. 91, no. 5, pp. 1118–1139, 2018.

[13] B. M. Kessels, "Ai-based model updating for nonlinear dynamical systems," PhD thesis, Eindhoven University of Technology, Eindhoven, The Netherlands, 2025.

[14] A. Retzler, R. Tóth, M. Schoukens, G. I. Beintema, J. Weigand, J.-P. Noël, Z. Kollár, and J. Swevers, "Learning-based augmentation of physics-based models: an industrial robot use case," *Data-Centric Engineering*, vol. 5, p. e12, 2024.

[15] R. Tóth, *Modeling and Identification of Linear Parameter-Varying Systems*, ser. Lecture Notes in Control and Information Sciences. Berlin, Heidelberg: Springer, 2010, vol. 403.

[16] R. Drenth, J. H. Hoekstra, M. Schoukens, and R. Tóth, "Efficient learning of affine and rational dependency LPV models with linear fractional representation," 2026.

[17] A. Ovchinnikov, A. Pillay, G. Pogudin, and T. Scanlon, "Computing all identifiable functions of parameters for ODE models," *Systems & Control Letters*, vol. 157, p. 105030, 2021.

[18] M. Gautier and W. Khalil, "Direct calculation of minimum set of inertial parameters of serial robots," *IEEE Transactions on Robotics and Automation*, vol. 6, no. 3, pp. 368–373, 1990.

[19] R. Drenth, "Efficient gradient-based learning of LPV models with linear fractional representation," Master's thesis, Eindhoven University of Technology, 2025.

[20] G. I. Beintema, M. Schoukens, and R. Tóth, "Deep subspace encoders for nonlinear system identification," *Automatica*, vol. 156, p. 111210, 2023.

[21] J. H. Hoekstra, B. M. Györök, R. Tóth, and M. Schoukens, "Encoder initialisation methods in the model augmentation setting," 2026.

[22] U. Forssell and L. Ljung, "Closed-loop identification revisited," *Automatica*, vol. 35, no. 7, pp. 1215–1241, 1999.

[23] A. Hefny, C. Downey, and G. J. Gordon, "Supervised learning for dynamical system learning," in *Advances in Neural Information Processing Systems*, vol. 28, 2015.

[24] C. Downey, A. Hefny, B. Li, B. Boots, and G. J. Gordon, "Predictive state recurrent neural networks," in *Advances in Neural Information Processing Systems*, vol. 30, 2017.

[25] B. M. Györök, J. H. Hoekstra, J. Kon, T. Péni, M. Schoukens, and R. Tóth, "Orthogonal projection-based regularization for efficient model augmentation," in *Proceedings of the 7th Annual Conference on Learning for Dynamics and Control*, ser. Proceedings of Machine Learning Research, N. Ozay, L. Balzano, D. Panagou, and A. Abate, Eds., vol. 283, 2025, pp. 166–178.

[26] C.-Y. Lee, S.-H. Hwang, E. Nam, and B.-K. Min, "Identification of mass and sliding friction parameters of machine tool feed drive using recursive least squares method," *The International Journal of Advanced Manufacturing Technology*, vol. 109, no. 9-12, pp. 2831–2844, 2020.

APPENDIX A
SUPPORTING DERIVATIONS

A. Explicit LPV-LFR realization

In (5), M_0 is (3) evaluated at $Y = 0$, and

$$M_1 = \begin{bmatrix} 0 & -m_h & 0 \\ -m_h & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}, \qquad M_2 = \begin{bmatrix} 0 & 0 & 0 \\ 0 & m_h & 0 \\ 0 & 0 & 0 \end{bmatrix}.$$

With a_1 and a_2 from (7) treated as independent signals, solving $M_0\ddot{q} + M_1 a_1 + M_2 a_2 = u - C\dot{q} - Kq$ for $\ddot{q}$ with $R = M_0^{-1}$ gives $\ddot{q} = R(u - Kq - C\dot{q} - M_1 a_1 - M_2 a_2)$. With $\dot{\tilde{x}} = \mathrm{col}(\dot{q}, \ddot{q})$, $z = \mathrm{col}(\ddot{q}, a_1)$, and $w = \mathrm{col}(a_1, a_2)$, the LTI part G of (10), partitioned as in [16, Eq. (6)], is

$$\begin{bmatrix} \dot{\tilde{x}} \\ z \\ q \end{bmatrix} = \underbrace{\begin{bmatrix} A_x & B_w & B_u \\ C_z & D_{zw} & D_{zu} \\ C_y & D_{yw} & D_{yu} \end{bmatrix}}_{G} \begin{bmatrix} \tilde{x} \\ w \\ u \end{bmatrix}. \tag{34}$$

with blocks

$$\begin{aligned} A_x &= \begin{bmatrix} 0 & I_3 \\ -RK & -RC \end{bmatrix}, & B_w &= \begin{bmatrix} 0 & 0 \\ -RM_1 & -RM_2 \end{bmatrix}, & B_u &= \begin{bmatrix} 0 \\ R \end{bmatrix}, \\ C_z &= \begin{bmatrix} -RK & -RC \\ 0 & 0 \end{bmatrix}, & D_{zw} &= \begin{bmatrix} -RM_1 & -RM_2 \\ I_3 & 0 \end{bmatrix}, & D_{zu} &= \begin{bmatrix} R \\ 0 \end{bmatrix}, \\ C_y &= \begin{bmatrix} I_3 & 0 \end{bmatrix}, & D_{yw} &= 0, & D_{yu} &= 0. \end{aligned} \tag{35}$$

Here $\tilde{x}, z, w \in \mathbb{R}^6$ and $u, q \in \mathbb{R}^3$, so $G \in \mathbb{R}^{15\times 15}$. The actuator frame enters only through $u = Pu_{\mathrm{act}}$ and $q_{\mathrm{act}} = P^\top q$ in (1).

Substituting $w = \Delta(Y)z$ into the z equation of (34) gives the algebraic loop equation $(I_6 - D_{zw}\Delta(Y))z = C_z x + D_{zu}u$. Written out,

$$\begin{bmatrix} I_3 + YRM_1 & YRM_2 \\ -YI_3 & I_3 \end{bmatrix} \begin{bmatrix} \ddot{q} \\ a_1 \end{bmatrix} = \begin{bmatrix} R(u - Kq - C\dot{q}) \\ 0 \end{bmatrix}.$$

The second block row gives $a_1 = Y\ddot{q}$, and the first then gives $M(Y)\ddot{q} = u - C\dot{q} - Kq$, that is (2), after multiplication by M_0. The condition of [16, Def. 1], that $I_6 - D_{zw}\Delta(Y)$ is nonsingular, is therefore the condition that $M(Y)$ is nonsingular: the Schur complement of the lower identity block gives

$$\begin{aligned} \det\big(I_6 - D_{zw}\Delta(Y)\big) &= \det\big(I_3 + YRM_1 + Y^2RM_2\big) \\ &= \det M(Y)/\det M_0. \end{aligned}$$

Since $M(Y)$ is nonsingular, the loop is solved in closed form, $\ddot{q} = M(Y)^{-1}(u - C\dot{q} - Kq)$ with

$$M(Y)^{-1} = \frac{N_0 + YN_1 + Y^2N_2}{d(Y)}, \qquad d(Y) = \det M(Y),$$

where $N_0 + YN_1 + Y^2N_2 = \mathrm{adj}\, M(Y)$ and $d(Y)$ are both quadratic in Y. The implementation evaluates both polynomials with Horner's scheme. [**TODO:** Missing: print N_0, N_1, N_2 and $d(Y)$ explicitly? Candidate: no, appendix budget; the quadratic degree was checked symbolically (sympy, 2026-10-06).]

TABLE III: Nominal baseline parameters.

Parameter	Value	Parameter	Value
m_1	10.2 kg	m_2	10.7 kg
m_b	22.8 kg	m_h	10.1 kg
J_b	1.00 kg m^2	J_h	0.05 kg m^2
c_{g1}	14.5 N s/m	c_{g2}	20.3 N s/m
c_y	10.0 N s/m	d	0.10 m
c_{b1}	9.0 N m s/rad	c_{b2}	9.0 N m s/rad
k_{b1}	1987.5 N m/rad	k_{b2}	1987.5 N m/rad
L_b	0.725 m		

B. Identifiable combinations

For $m_h > 0$, the ten distinct entries of M_0, C and K determine θ_{base} of (11) uniquely (Section II-C):

$$\begin{aligned} &m_h = M_{0,33}, && d = -M_{0,23}/m_h, && m_\Sigma = M_{0,11} - m_h, \\ &m_\Delta = 2M_{0,12}/L_b, && J_{\mathrm{eff}} = M_{0,22} - m_h d^2, && k_{b,\Sigma} = K_{22}, \\ &c_{g1} = \tfrac{1}{2}C_{11} + C_{12}/L_b, && c_{g2} = \tfrac{1}{2}C_{11} - C_{12}/L_b, && c_y = C_{33}, \\ &c_{b,\Sigma} = C_{22} - \tfrac{L_b^2}{4}C_{11}. \end{aligned}$$

Conversely, (3) and (4) give every entry as a function of θ_{base}, so the map is one-to-one.

[**TODO:** Remarks cut from Section II-B; keep here or drop?]

- Y^2 is realized by repeating Y, not as a second scheduling variable $p_2 = Y^2$, which would admit pairs with $p_2 \neq Y^2$; extra scheduling variables cause overbounding in controller synthesis [16]
- The scheduling map is a known selection of the state ($Y = q_3$), not a learned map as in the self-scheduled models of [16]
- The exact condition $\det M(Y) \neq 0$ replaces the sufficient condition of [16, Thm. 6], which needs a scheduling set in the unit ball and $\rho(D_{zw}) < 1$
- Drenth's Def. 1 is stated in discrete time but constrains only the algebraic loop equation, so it applies unchanged in continuous time

C. Nominal physical parameters

Table III lists the nominal values used by the baseline. They are taken from the ETEL gantry reported by García-Herreros et al. [3, Table I].

D. Positive definiteness of the inertia matrix

With $a = L_b m_\Delta/2$, elimination of the m_h block and completion of the square give

$$\begin{aligned} \frac{\det M(Y)}{m_h} &= (m_\Sigma + m_h)(J_{\mathrm{eff}} + m_h Y^2) - (a - m_h Y)^2 \\ &= m_\Sigma m_h \left(Y + \frac{a}{m_\Sigma}\right)^2 \\ &\quad + (m_\Sigma + m_h)\left(J_{\mathrm{eff}} - \frac{a^2}{m_\Sigma}\right). \end{aligned}$$

The 2×2 Schur complement has positive leading entry $m_\Sigma + m_h$. Sylvester's criterion therefore reduces positive definiteness to positivity of the completed-square expression for every Y. Hence, for positive m_h, m_Σ, and J_{eff}, $M(Y) \succ 0$ for all $Y \in \mathbb{R}$

if and only if (12) holds. For the physical parameterization, the difference between the two sides is

$$m_\Sigma J_{\text{eff}} - \frac{L_b^2}{4} m_\Delta^2 = m_\Sigma (J_b + J_h) + \frac{L_b^2}{4} \left[4 m_1 m_2 + m_b (m_1 + m_2) \right],$$

which is strictly positive for positive component masses and yaw inertias.

In (23), let $b(\theta_{\text{base}}) = \rho (2/L_b) \sqrt{m_\Sigma J_{\text{eff}}}$. The offset and scale are

$$r_0 = \frac{m_{\Delta,0}}{b(\theta_{\text{base},0})}, \quad \eta_0 = \operatorname{atanh}(r_0), \quad s = \frac{m_{\Delta,0}}{b(\theta_{\text{base},0})(1 - r_0^2)}.$$

Here η_0 makes $\xi_\Delta = 0$ reproduce the initial combination, and s gives $\partial m_\Delta / \partial \xi_\Delta |_0 = m_{\Delta,0}$, matching the local relative-change scaling of the logarithmic coordinates. The construction requires $m_{\Delta,0} \neq 0$; a start near zero makes s small and the coordinate numerically inert. The exact inverse used for reporting is

$$\xi_\Delta = \frac{\operatorname{atanh}(m_\Delta / b(\theta_{\text{base}})) - \eta_0}{s}.$$

E. PS stopping rule

The unexplained fraction ν of (27) is evaluated after every i_cth update of V, with ν_i its value after update i. S1 ends at update

$$i_1 = \min \big\{ i \in \mathcal{J} : \; i = i_{\max} \text{ or } \big(i \geq i_{\min} \text{ and } \nu_i^{\text{new}} > (1 - \delta_\nu)\, \nu_i^{\text{old}} \big) \big\},$$

$$\nu_i^{\text{new}} = \min_{j \in \mathcal{W}_i} \nu_j, \qquad \nu_i^{\text{old}} = \min_{j \in \mathcal{J}_i \setminus \mathcal{W}_i} \nu_j,$$

where $\mathcal{J} = \{i_c, 2i_c, \dots\}$, $\mathcal{J}_i = \{j \in \mathcal{J} : j \leq i\}$, $\mathcal{W}_i$ the last w elements of $\mathcal{J}_i$, δ_ν the plateau tolerance and $i_{\min}$, $i_{\max}$ the earliest and latest end of S1, with $\nu_i^{\text{old}} = \infty$ while $\mathcal{J}_i$ has at most w elements. S2 runs if $\nu_{i_1} \leq \nu_{\max}$ and, when $i_1 > i_s$, also $\nu_{i_s} \leq \nu_{\max}$, where $i_s < i_{\max}$ is a fixed early check. If $\nu_{i_s} > \nu_{\max}$, PS ends at i_s. The values are given in Section V.

F. OBC and experiment details

- Derive the stacked projector, rank-deficient pseudoinverse case, epoch refresh, and additional-state zero extension
- Derive $J^+ \Delta^*$ and list the measured singular-direction contributions to the predicted parameter bias
- List dataset identifiers, split membership, hyperparameters, seed, and exact metric definitions for every reported arm
- Figure (existing, optional): place `controller-chain-v1` here only if the controller reconstruction cannot be audited adequately in prose and equations

[**TODO:** Can code and data tables be cited through a repository or DOI instead of spending paper space on full reproducibility metadata?]